One Concept, Multiple Geometries

A spelling feature fakes a circle of fifths, and next-key statistics, not co-occurrence, predict what language models do with musical keys

Jan Kirin

github.com/VykosMolt | github.com/VykosMolt/One-Concept-Multiple-Geometries | ORCID 0009-0005-2681-9027

Corrected preprint. Every number traces to a results file in the repository; `V4_CORRECTION_REPORT.md` records the corrected Phase-V baseline and its consequences, while the chronological logs distinguish fixed from post-hoc analyses.

Abstract

Studies of representation geometry treat the shape of a concept family as a fact about the domain: weekdays form a circle, numbers a line. We show on a natural domain that the shape depends on which statistic is computed and which task is measured. Musical keys carry two legitimate geometries: a circle of fifths when enharmonic spellings such as C♭ and B are identified, and an open line of fifths when they are kept apart. Co-occurrence statistics of Wikipedia recover the circle, and a naive Fourier analysis of key-name activations in four language models seems to agree. The neural harmonic is a spelling artefact: a binary "has an accidental" feature places 80% of its Fourier energy in the fifths pair, and respelling white keys shows that what survives follows the glyph, though the respelled names are also rare. The models' next-key behaviour instead follows the open line, and the corpus statistic that matches it is not co-occurrence but the ordered conditional, how often key j follows key i (Spearman 0.80 against 0.1–0.54 for PMI). Whether that conditional predicts held-out behaviour beyond a rich tonal and spelling baseline is answered only weakly: a small residual that rests largely on one rare key. The geometry one reads off a concept family belongs to the statistic, the equivalence relation and the task, not to the family alone.

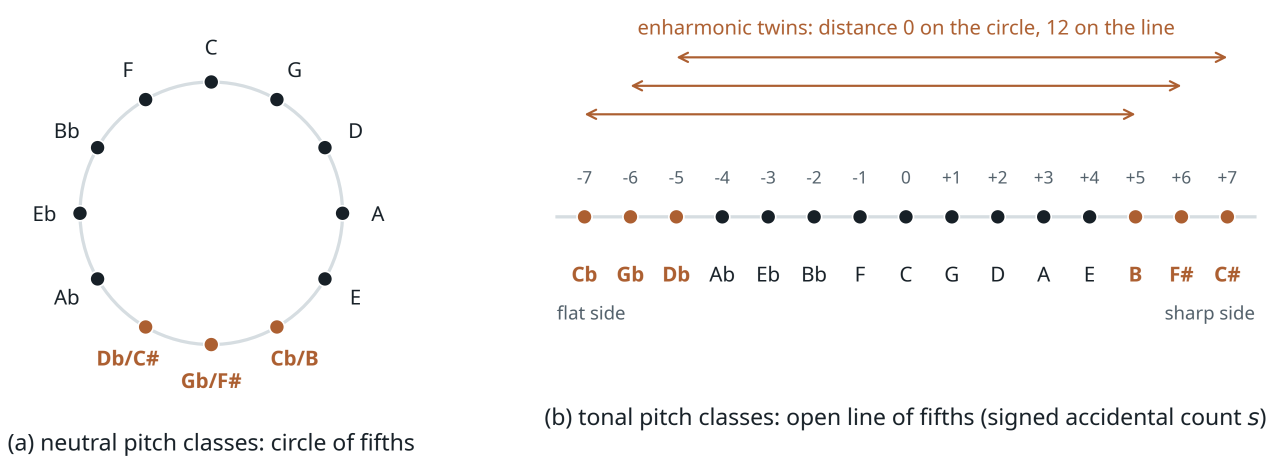

Figure 1. The two tonal spaces used throughout. (a) Neutral pitch classes identify enharmonic spellings and close into the circle of fifths. (b) Tonal pitch classes keep spellings distinct on an open line indexed by the signed accidental count s of the key signature (-7 = seven flats, $+7$ = seven sharps); the three standard enharmonic pairs (orange) are distance 0 on the circle and 12 on the line. Over the 105 pairs of the 15 standard major-key spellings the two distance matrices correlate only $\rho = 0.25$ (vs. 0.68 for the conventional 12-key set), which is what makes the 15-key design discriminative.

Results at a glance

Question. Is there one "geometry of musical keys" to be read off language statistics or language-model activations? And if different measurements disagree, which measurement predicts what?

What is established. ("$X \mid Y$" denotes the partial Spearman correlation of a distance matrix with geometry X controlling for geometry Y and for orthographic and commonness features, §2; ECI is the enharmonic collapse index defined in §3.)

- *Symmetric corpus statistics are periodic.* Wikipedia PMI between keys falls with fifths distance (partial correlation +0.62 after spelling and frequency controls); the circle beats the line (§3).

- *The neural circle is spelling.* An accidental-glyph feature carries 80% of its Fourier energy in the fifths pair, and respelling white keys shows the block follows the glyph; after controls only the 7B model keeps any fifths signal, and it is line-like (§4).
- *Behaviour follows the open line.* Asked which key comes next, all four models order the keys by signed accidental count (§5).
- *The next-key table matches behaviour; PMI does not.* Spearman $\approx$ 0.80 against 0.1–0.54; PMI identifies enharmonic twins, the table does not; direction is irrelevant on a co-mention matrix this symmetric (§6).
- *Two artifact explanations fail in synthetic models.* Neither a line-aligned output code nor rare-alias learning produces the line on its own (Appendix E).
- *Beyond theory geometry the residual is small and fragile.* The conditional cuts held-out KL by 6–13% in three of four models, but most of that is one rare key, C♭ major (Appendix F).
- *Other corpora and one training run do not extend it.* Gains vary in sign across corpora, the corpus-specificity test is directionless, and the checkpoint residual has no stable onset: no training-data fingerprint (Appendix F).

What is not claimed. That LLMs "contain" or "reason with" a circle of fifths; a circle-to-line transition across layers; a privileged predicting-position locus; a scaling trend; separable circle/line/spelling subspaces; a tokenization or rare-alias explanation of the line; or that behaviour fingerprints a model's training corpus (nor that it does not). Each of these was proposed during the project and killed by a control, a null or an independent review (Appendix A).

How to read this paper

§3–§4 treat the symmetric statistic and the instrument failure; §5–§6 contain the main result; §7 summarizes the two follow-ups that Appendices E and F report in full. Appendix A lists every retracted claim and post-hoc decision, Appendix B the Fourier bookkeeping, Appendix C the tables, Appendix D the methods.

1 Introduction

A growing body of work reads geometric structure off the internal representations of language models — circles for weekdays and months (Engels et al., 2025), lines and helices for numbers (Gurnee and Tegmark, 2024; Bassi and Tomar, 2026), lattices for grids (Brenner et al., 2026) — and explains it from the statistics of the training text. The sharpest version of that explanation is due to Karkada et al. (2026): when a word–word association statistic is translation-symmetric over a concept family, word2vec-like models embed the family on a Fourier lattice whose harmonics are set by the statistic's spectrum, and transformer residual streams show the same qualitative picture. A second tradition (Zhao et al., 2024; Zhao and Thrampoulidis, 2025) starts from a different object, the conditional distribution of the next token given the context, which is what next-token prediction actually optimizes.

Co-occurrence and conditionals are different statistics of the same text. For most concept families studied so far they happen to agree, because the family has one obvious ordering. Musical keys do not. Twelve pitch classes carry two competing cyclic orderings, chromatic ($x \mapsto x + 1$) and fifths ($x \mapsto x + 7$), related by an automorphism of $\mathbb{Z}_{12}$; more importantly, music theory itself keeps two spaces for the same keys. In the *neutral* pitch-class space of pitch-class set theory, C♭ and B are the same object and the keys close into the circle of fifths. In the *tonal* pitch-class space of Temperley (2000) and Moss et al. (2023), spellings are kept, keys sit on an open line indexed by key-signature accidentals, and C♭ major is twelve steps from B major (Figure 1). Neither space is more correct; they are different equivalence relations, and — the point of Moss et al. (2023) — a neutral, MIDI-like encoding of the data throws the distinction away before any analysis begins.

That makes keys a good instrument. We ask which geometry shows up under which corpus statistic and which model measurement, which statistic matches the models' behaviour, and, as a bounded follow-up, whether that statistic still predicts held-out behaviour once the obvious geometric and spelling explanations are removed. The project began as a cheap test of the symmetric-statistics theory on a family with competing orderings. Its first result was a convincing false positive: the fifths harmonic in key-name activations turned out to be a spelling feature. What survived several rounds of controls, and two synthetic attempts to explain the natural result away, is a different and, we think, more general claim:

Central claim

Different statistics over the same concept family expose different, equally legitimate geometries. For keys, PMI-type co-occurrence identifies enharmonic spellings with each other and is periodic in fifths; the conditional tables built from the same local counts do not identify the spellings, whether or not the table is symmetrized, while staying fifths-smooth. The models' next-key behaviour lives on the spelling-sensitive open line and matches the conditional tables far better than any co-occurrence statistic we tried. A naive Fourier analysis of key-name activations reports a circle instead, because a spelling feature aliases into the fifths harmonic. A geometric reading of a concept family is therefore meaningful only relative to the statistic, the equivalence relation, the measurement and the task that define it. Whether the conditionals carry anything beyond a rich tonal and spelling baseline is answered only weakly: a small held-out residual that passes a relabeling test in three of four models but rests largely on a single held-out key in two of them, with a distributed effect only in Gemma, and no cross-corpus or training-data specificity (Appendix F).

What is not new. Circle-of-fifths geometry in models trained on *music* is established (Chuan et al., 2020; Huang et al., 2016; Sadek and Bakarji, 2026); the line of fifths as tonal structure goes back to Weber's 1821 spiral, straightened by Temperley (2000); the coexistence of string and semantic geometry (Marjieh et al., 2025), context-dependent concept geometry (Hu et al., 2026), transient mid-layer perceptual geometry (Singh and Chopra, 2026), correlation-driven feature geometry (Prieto et al., 2026), factored representations (Shai et al., 2026), and the correspondence of activation geometry with behaviour (Wurgaft et al., 2026) are all known; the DFT of the diatonic/black-key set peaking at the fifth coefficient is textbook in mathematical music theory (Quinn, 2006; Amiot, 2016); a Fourier-looking representation need not imply a Fourier algorithm (Nanda et al., 2023; Zhong et al., 2023; Feucht et al., 2026); and Fourier sparsity is necessary but not sufficient for linear separability (Fu et al., 2026). The conjunction — a natural family with two motivated geometries, operator-dependent recovery, a bounded and, as it turns out, fragile held-out behavioural correspondence with the task-aligned operator, and a realistic demonstration of the orthographic false positive — is what we claim.

2 Setting

2.1 Keys, coordinates and geometries

Twelve pitch classes carry two cyclic orderings, chromatic ($x \mapsto x+1$) and fifths ($x \mapsto x+7$). Because $7^2 \equiv 1 \pmod{12}$, the map $x \mapsto 7x$ swaps the two, so a statistic that is smooth in fifths is also circulant in semitones: circulant structure alone cannot say which ordering it follows, only the Fourier index can. We report paired Fourier energies $P_m = E_m + E_{12-m}$; P_1 is the chromatic fundamental and P_5 the fifths fundamental (definitions and checks in Appendix B).

The **12-key design** uses the conventional spellings C, D♭, D, E♭, E, F, F♯, G, A♭, A, B♭, B. The **15-key design** uses the fifteen standard major-key spellings on the signed line $s = -7$ (C♭) … $+7$ (C♯), so that (C♭, B), (G♭, F♯) and (D♭, C♯) share a pitch class: circle distance 0, line distance 12. Candidate geometries over the 105 pairs are the circle, the line, the chromatic cycle, an orthographic family (glyph class, accidental, edit distance, root letter, alphabet distance, token count) and log commonness. Classical tonal spaces add nothing on major keys alone (Krumhansl–Kessler similarity is $\rho = 0.99$ with the circle, Chew's spiral array $\rho = 1.00$ with the line; Krumhansl and Kessler, 1982; Chew, 2000), so we report circle against line. Corpus kernels and key-name geometry use the 12-key design; behaviour, operators and held-out tests use the 15-key design.

2.2 Models, corpus and behavioural task

Models: OLMo-2-0425-1B, Gemma-2-2B, Qwen2.5-3B and OLMo-2-1124-7B, base checkpoints (OLMo Team et al., 2025; Gemma Team et al., 2024; Qwen Team et al., 2024), forward passes only. Lists of four numbers follow this order (OLMo-1B / Gemma / Qwen / OLMo-7B). Corpus: English Wikipedia 20231101 (6.41M documents, $\approx$ 3.1B words), the dump used by Karkada et al. (2026), scanned with their $L = 16$ window for symmetric statistics and with ordered key-mention windows for conditionals (§6). Behaviour: for each source key and each of four relational context families (enharmonic, harmonic relation, chord progression, modulation), four prompt templates ask which key comes next, and we record the sequence log-probability of every candidate continuation "j major" over the 15 spellings, row-normalized. This is restricted-candidate behaviour, not free generation (Appendix D.4). The total log-probability scorer is primary; a length-normalized scorer and an enharmonic-merged scorer were pre-registered as checks. Appendix F also uses a 12-class target-aggregated view, a projection of the scored distribution onto pitch classes, chosen as the lead view for that analysis after both views were seen.

2.3 Statistics and nulls

Geometry is compared by representational similarity analysis (Spearman correlation between two matrices of pairwise distances) and by the partial correlations line | circle and circle | line, each controlling for the other and for the orthographic and commonness features (the exact control set of each analysis is in Appendix D.7). Every partial carries a key-relabeling null, and a glyph-preserving one where a spelling block is a plausible nuisance; best-over-layers values carry a max-over-layers null; corpus statistics carry a shard-cluster bootstrap and the held-out gain a document-cluster bootstrap. With 12–15 concepts the pairs are not independent, and we never quote asymptotic p-values.

3 Symmetric corpus statistics recover a periodic fifths structure

Months are the positive control: the Karkada statistic M^*, a normalized co-occurrence excess defined in Appendix B, is 97% circulant over Wikipedia and OLMo-2-1B's month-name Fourier profile matches its spectrum (best-layer cosine 0.97–0.99). For keys M^* saturates, because $P_{ij} \gg P_i P_j$ drives every off-diagonal entry to its ceiling and leaves a spectrum that flips with corpus size, so we use PMI, to which skip-gram embeddings are related through the factorization of shifted PMI (Levy and Goldberg, 2014). The PMI kernel is jagged in semitone order and falls smoothly with fifths distance (Figure 2b); its naive Fourier share $P_5 = 0.50$ depends on the diagonal, so the evidence is the kernel and the controlled partials, not the spectrum.

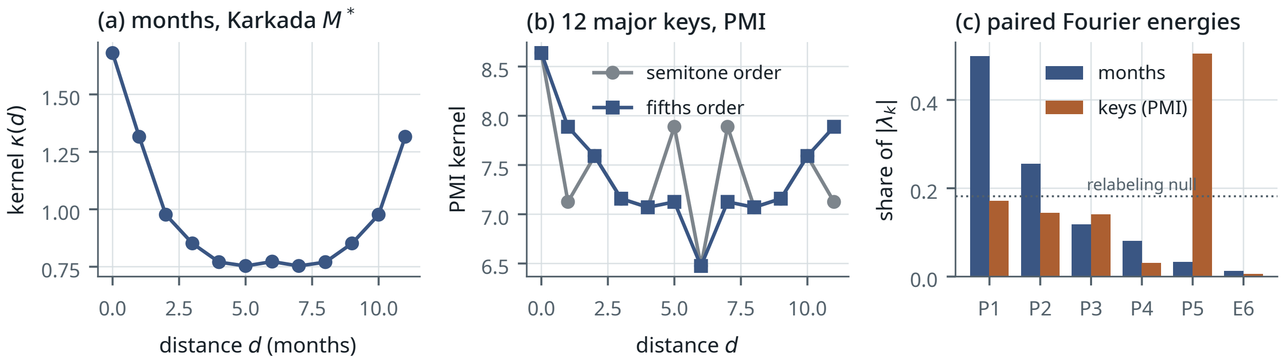

Figure 2. Symmetric corpus statistics. (a) The months kernel (positive control). (b) The keys PMI kernel is jagged in semitone order and smooth in fifths order. (c) Paired Fourier shares of $|\lambda_k|$ (months from M^*, keys from PMI; M^* is saturated for keys and its key spectrum is not shown); the keys' P_5 share is diagonal-dependent and is not the evidence — the controlled partials are.

Result 1 — the corpus fifths structure is real, ordering-specific and periodic

Wikipedia's PMI between keys falls with fifths distance: the partial correlation with fifths distance, after the spelling and frequency controls, is +0.62 with canonical spellings and +0.66 with enharmonics merged, at the resolution of the 2,000-relabeling null. The structure is periodic, not open: with enharmonics merged, circle | line is +0.58 and line | circle is −0.07, and a shard-cluster bootstrap puts the circle above the line in every draw (Table 1). It lives in the surrounding vocabulary, as Karkada et al. predict: a helper-word factorization of the 3000-word PMI reproduces the key rows' fifths partial at +0.52...+0.56, and deleting every document that mentions the circle of fifths changes nothing (+0.64). Two cautions sit in the table notes: canonical spellings starve the enharmonic seam, so we lead with the merged family, and the partial was still rising with corpus size at 3.1B words.

Table 1. Corpus fifths structure over the 12 major keys (Wikipedia PMI; 66 pairs). Partial Spearman correlations after controlling for the accidental block, log commonness, letter identity and alphabet distance; brackets are shard-cluster bootstrap 95% intervals (41 shards, 1,000 draws). "Circle beats line" is the bootstrap probability that circle | line exceeds line | circle.

Statistic	canonical spellings	enharmonics merged
fifths partial	+0.62 [+0.44, +0.68]	+0.66 [+0.49, +0.71]
circle | line	+0.40 [+0.05, +0.55]	+0.58 [+0.39, +0.64]
line | circle	+0.27 [−0.08, +0.58]	−0.07 [−0.20, +0.07]
bootstrap P(circle beats line)	0.63	1.00
white-key order F–C–G–D–A–E–B	rank 78 of 5,040 ($p = 0.016$)	—
helper-word factorization, fifths partial	+0.52...+0.56	—

Canonical spellings starve the seam pairs because Wikipedia spells consistently within a document (D♭|F♯ weighted count 17 against D♭|G♭ 160). Across corpus-size prefixes the canonical fifths partial runs +0.27, +0.24, +0.35, +0.43, +0.45, +0.63 from 0.11B to 3.12B words, so its asymptotic size is not established. An earlier Poisson-on-counts bootstrap ignored document clustering and is withdrawn (Appendix A).

Over the 15 spellings the same statistic is both periodic and open. The three enharmonic pairs are the closest pairs of all: the enharmonic collapse index (ECI), the mean percentile rank of the three twin distances among all pairs, is 0.04, where 0 means closest of all and 0.5 means unremarkable (the bootstrap interval reaches 0.44 because C♭ major has 38 mentions). At the same time the controlled partials are circle | line +0.42 and line | circle +0.38. The neutral identity at the seam is mostly talk about notation: Wikipedia contains only 41 co-mentions of an enharmonic pair, and 20 of them contain the word "enharmonic".

4 Naive key-name geometry is an orthographic alias

Key-name activations of all four models show a fifths Fourier share P_5 of 0.26–0.55 with the chromatic share at the null, in every template, position and spelling variant. This is exactly what the months instrument reports when applied verbatim, and the first draft of this project called it a circle of fifths. It is not one.

Result 2 — the fifths harmonic is an accidental-glyph feature

The five conventionally spelled keys with an accidental sit next to each other on the circle of fifths, so a binary "has an accidental" feature is a five-wide block in fifths coordinates and puts 79.6% of its Fourier energy into the fifths pair (Figure 3a). That is enough: projecting the feature out drops the models' P_5 to the relabeling null (0.11–0.14 against 0.12), and the $k = 7$ mode is a single direction rather than a circle (isotropy 0.4–0.65, against 0.98 for a true circle). Respelling settles what the block tracks. Writing four white keys as B♯, F♭, E♯, C♭ separates "carries an accidental glyph" from "is a black key"; after respelling, the black-key RSA collapses to about zero in all four models while the glyph RSA keeps a third to four fifths of its strength (+0.51/+0.43/+0.67/+0.28 at the best of six layers; Figure 3b), with the caveat that the respelled names are also a thousand times rarer in the corpus. The failure mode is general: a quarter of all binary partitions of the 12-cycle put more than half their Fourier energy into a single paired mode, and 5% into the fifths pair specifically (Appendix B).

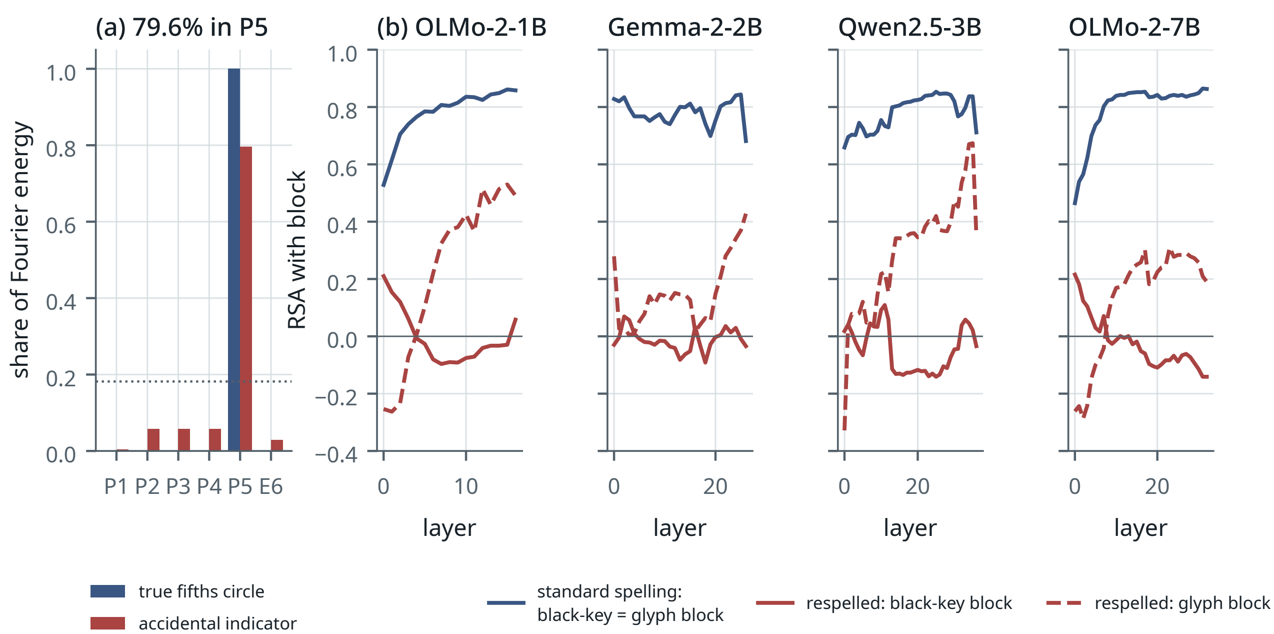

Figure 3. The false neural circle. (a) Paired Fourier shares of a true fifths circle and of the binary accidental indicator: the indicator alone is a convincing "P_5 signal". (b) Respelling intervention, per layer: with standard spellings the black-key and accidental-glyph blocks coincide (blue); after respelling C, E, F, B as B♯, F♭, E♯, C♭ the block follows the glyph (dashed red), not the black keys (solid red). The respelled strings are rare, so glyph and rarity remain entangled; key-signature membership is ruled out.

What remains after controls is small. With the glyph block and the frequency and letter features partialled out, only the 7B model keeps a fifths partial that survives a max-over-layers correction (+0.34, $p = .046$), and even that signal is line-like and gone at the final layer; the other three models are at noise. Two further instrument facts are worked out in Appendix G: the last token of a multi-token key name is not a neutral probe (it produced a line in Qwen and 7B that vanishes under a span-mean probe), and at the one token-uniform position spelling and commonness explain much of the pair geometry, with circle or line adding almost nothing. Hidden-state geometry is therefore not where this paper's positive claim lives.

5 Next-key behaviour follows the open line

Result 3 — the behavioural space is Temperley's line

Asked which key comes next, the models order the keys along the open line. The line partial line | circle is +0.14…+0.56 across models and context families, with all templates significant under the glyph-preserving null in 15 of 16 model × family cells, while the circle partial is near zero in the three smaller models and +0.13…+0.28 in the 7B (Table C.6). The first principal axis of the predictive matrices tracks the signed accidental count ($|\rho| = 0.70$–0.80), and the top answers in relational families cluster one line step away, at the dominant or the subdominant. The line survives the pre-registered length-normalized scorer in every cell. It does not survive merging the enharmonic twins before scoring, in the three families where that scorer applies (line | circle −0.23…−0.01); the older 12-key merged-target analysis keeps a positive line component (line | circle +0.69 → +0.46, +0.74 → +0.52, +0.79 → +0.66, +0.74 → +0.56), and the two experiments must not be conflated.

Enharmonic identity itself, the defining property of the neutral space, is known behaviourally by the 7B model only: asked for "the enharmonic equivalent of X major" it gives ECI 0.11, Qwen 0.51, and OLMo-1B and Gemma no collapse at all (0.70 and 0.83); outside that family every model keeps twins far apart. Few-shot fifth-relation accuracy rises with model size (dominant 0.08/0.33/0.75/0.92), but with four models from four families that is not a scaling trend, and the geometric statements above show no such ordering.

So the corpus's symmetric statistic is periodic and the models' behaviour is open. Phase II concluded from the cue-conditioned symmetric statistics that "the corpus contains no task-conditioned line". That conclusion rested on a convention: it booked spelling as noise. Under Temperley (2000) and Moss et al. (2023) the signed accidental count of the key signature *is* the tonal coordinate, and in text, spelling class and that coordinate are the same variable. Which convention is right is a music-theoretic question the data cannot settle. The next section replaces it with a question they can.

6 The next-key table matches behaviour; PMI does not

Instead of asking whether the corpus "is" a circle or a line, we ask which corpus statistic predicts which model observable. Three are computed on the same Wikipedia text over the 15 spellings: *SYM*, Karkada's symmetric PMI over an $L = 16$ window; *HELP*, the helper-word factorization; and *COND*, the ordered conditional $p(j \mid i)$, the rows of the 15×15 count matrix N of mentions of key j within 40 words after key i, compared by Jensen–Shannon divergence between rows. Because SYM and COND differ in several ways at once, three matched operators are built from the same N: a symmetrized conditional (rows of $N + N^{\top}$), a PMI from the same counts, and the reverse conditional (rows of $N^{\top}$); all were fixed before the comparison was run (Appendix D.3). One fact about N matters for reading the control: the 40-word co-mention counts are almost order-symmetric (Pearson correlation between N and $N^{\top}$ of 0.977), so on this corpus a directional and a symmetrized conditional cannot differ much. In the vocabulary of distributional semantics, PMI measures first-order association between two keys, while the conditional rows compare the second-order contexts that follow each key (Schütze and Pedersen, 1993; Sahlgren, 2008); the question is which of the two the models' behaviour follows.

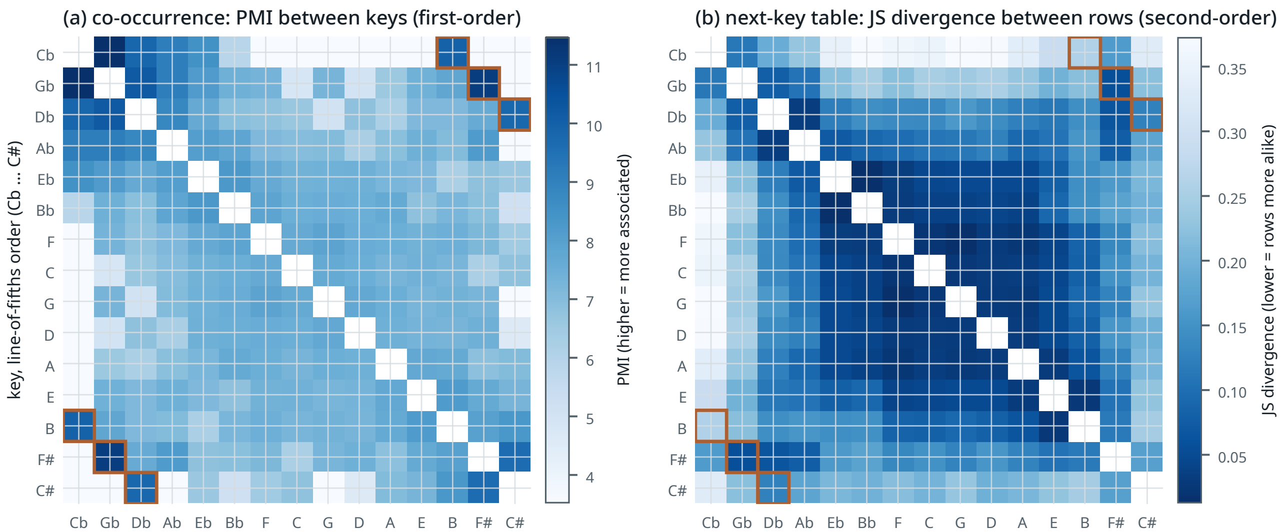

Figure 4. Two statistics of the same text, over the 15 spellings in line-of-fifths order. (a) PMI between keys (Karkada window, $L = 16$): darker is more associated. The three enharmonic twin pairs (orange boxes) are among the most associated cells of all (ranks 2, 4 and 6 of 105). (b) The next-key table: Jensen–Shannon divergence between the rows $p(\cdot \mid i)$ of 40-word co-mention counts, darker is more alike. The twins are not identified: their rows are no more alike than a typical pair (ECI 0.57), and a dark band runs along the line of fifths. Both matrices are fifths-smooth; they disagree only about whether C♭ and B are the same key.

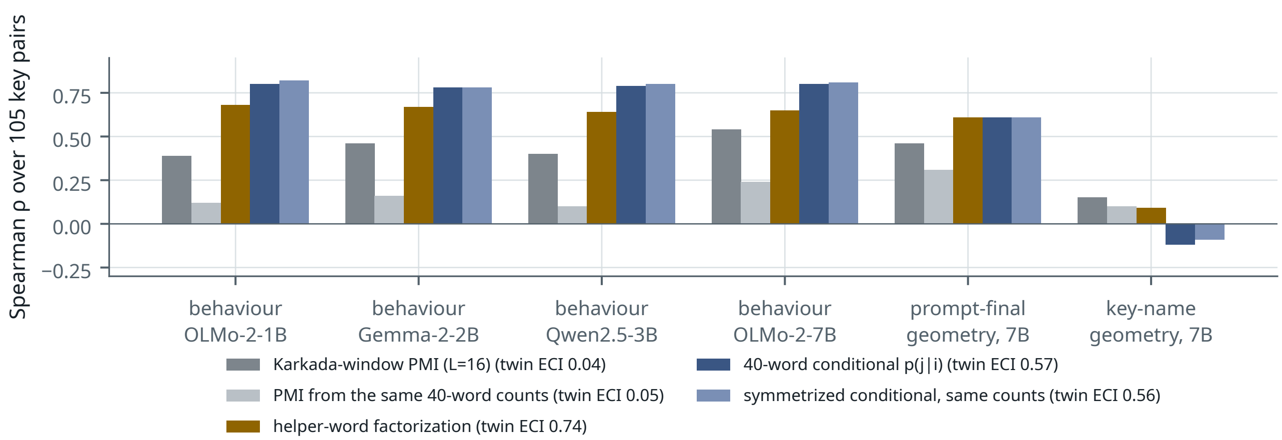

Figure 5. Which operator predicts which observable. Spearman correlation between each operator's pairwise distances and those of the observable (restricted next-key behaviour in the modulation family, four models; 7B prompt-final geometry at layer 18 and key-name span-mean geometry at layer 24 — fixed layers from Phase II, with no max-over-layers null applied here). First-order operators (greys/gold): Karkada-window PMI, a PMI built from the same 40-word counts as the conditionals, and the helper-word factorization. Second-order operators (blues): the directional 40-word conditional and a symmetrized conditional from the same counts. Twin ECI in the legend: mean percentile rank of the three enharmonic pairs (0 = closest, 0.5 = null).

Result 4 — the contrast is conditional-row vs association, not directional vs symmetric

The two kinds of statistic disagree about the enharmonic twins and agree about everything else (Table 2). PMI, however it is built, puts the twins closest of all 105 pairs (ECI 0.04–0.05); the conditional rows do not identify them (ECI 0.56–0.60; two of the three bootstrap intervals include chance and the third starts at 0.53), and a count matrix with the same margins and no structure would put them even farther apart (0.84), so the rows place twins closer than noise would while still not merging them. Both kinds remain fifths-periodic under control. On behaviour the ordering is clear: the conditional rows correlate with the models' next-key distributions at Spearman 0.78–0.83, the helper-word factorization at 0.64–0.68, Karkada's PMI at 0.39–0.54, and a PMI built from the very same 40-word counts at 0.10–0.24 (Figure 5). The three conditional rows are near-identical objects ($\rho = 0.97$–0.99 between their distance matrices), so their agreement is expected; the informative contrast is between them and the PMI built from the same counts. No operator predicts key-name hidden-state geometry well (nothing above +0.15 except the helper factorization on Qwen at +0.37).

Table 2. Corpus operators over the 15 spellings. Twin ECI: mean percentile rank of the three enharmonic pairs (0 closest of all, 0.5 unremarkable), with bootstrap 95% intervals where available (shard-cluster for the Karkada window, `results/corpus/wiki/cluster_boot.txt`; document-cluster for the 40-word operators, `results/phase5/operator_eci_boot_v4.txt`). Behaviour: Spearman correlation with the models' next-key distributions in the modulation family (OLMo-1B / Gemma / Qwen / OLMo-7B).

Operator	kind	twin ECI	Spearman with behaviour
Karkada-window PMI ($L = 16$)	first-order	0.04 [0.03, 0.44]	+0.39/+0.46/+0.40/+0.54
PMI from the 40-word counts	first-order	0.05 [0.04, 0.27]	+0.12/+0.16/+0.10/+0.24
helper-word factorization	first-order	0.74	+0.68/+0.67/+0.64/+0.65
40-word conditional $p(j \mid i)$	second-order	0.57 [0.47, 0.67]	+0.80/+0.78/+0.79/+0.80
symmetrized conditional, same counts	second-order	0.56 [0.49, 0.68]	+0.82/+0.78/+0.80/+0.81
reverse conditional, same counts	second-order	0.60 [0.53, 0.71]	+0.83/+0.78/+0.80/+0.81
marginals-only null matrix	—	0.84 ± 0.05	—

This is the descriptive statement of operator-dependent geometry. What separates the operators that match behaviour from those that do not is how they are built: a table of what follows each key, compared row by row, against a measure of how often two keys co-occur relative to chance. On this corpus the direction of the conditional cannot matter because the counts are nearly symmetric. The dichotomy is not sharp: the helper-word factorization, an association operator and the one Karkada's theory is about, lands between the two groups and keeps twins farthest of all. The local conditional is the operator closest to the restricted next-key task among those tested; it is not the next-token objective itself, and this comparison does not test either factorization theory. Used as held-out regressors, the matched operators separate less cleanly (Appendix F.1).

7 Two follow-ups, in brief

Two further studies are reported in full in Appendices E and F. Their conclusions are negative or bounded, so we summarize them here rather than let them carry the paper.

Synthetic controls (Appendix E). Two cheap explanations of the natural line were tested in small transformers trained from scratch on synthetic data with no music vocabulary. A line-aligned output code over a perfectly periodic law leaves converged behaviour periodic while tilting the predictive-state representation toward the line by about +0.1 (Result 5), and rare source aliases are learned exactly as fast as any other rare state (Result 6). In this toy world neither spelled outputs nor rarity produces a line on its own; other explanations are not excluded.

Beyond correspondence (Appendix F). Result 4 could still mean only that both matrices look like the line, so we asked whether the corpus conditional predicts held-out behaviour after true fifths-circle, chromatic, open-line, spelling, frequency and tokenizer features are removed. The answer is a small and fragile residual: the window conditional reduces held-out KL by 6%, 13% and 7% in OLMo-1B, Gemma and OLMo-7B and not at all in Qwen, but one held-out row, C♭ major, supplies two thirds, half and almost all of those three gains, only Gemma's survives dropping it, and nothing survives a correction over the 288 tests run (Result 7). Other corpora give heterogeneous gains and a directionless specificity test, so there is no training-data fingerprint (Result 8), and along OLMo-2-1B checkpoints the line appears by 21B tokens while the residual has no stable onset (Result 9).

8 Discussion

One concept, several geometries. There is no context-free geometry of musical keys waiting to be read off language statistics or activations. PMI-type co-occurrence identifies enharmonic twins with each other and carries a robust periodic fifths structure; Wikipedia shows it, and a Karkada-style helper-vocabulary factorization reproduces it. Conditional tables do not merge spelled tonal states, because C♭ major and B major occur in different contexts and are followed by different keys, and it is these tables, symmetrized or not, that match the models' next-key behaviour. The matched control shows that the difference comes from how the statistic is built, not from its direction, though on a co-mention matrix this symmetric the two could hardly differ; in the vocabulary of distributional semantics the twins are bound at first order and distinct at second order, and behaviour follows the second. Hidden-state geometry is dominated by lexical structure that projects onto exactly the harmonics used to diagnose semantic symmetry. The corrected held-out analysis adds a narrower result: some source-specific conditional structure remains beyond a rich baseline in three models by a relabeling test, but in two of them it is mostly one held-out key, Qwen shows none, and the bootstrap and cross-corpus results do not support a general claim. One alternative reading of the held-out residual is not excluded: the corpus conditional and the models' next-key behaviour may covary because both are downstream of the same small stock of textbook facts about key relationships, stated in similar words across many corpora, rather than because the models' representations track local co-occurrence structure. Nothing in Appendix F separates these, and the cross-corpus null is compatible with both. A geometric interpretation is meaningful only relative to the statistic, equivalence relation, measurement and task that define it.

Why orthography is not simply nuisance. Under Temperley (2000) and Moss et al. (2023) spelling is part of tonal semantics; treating it as a covariate reveals a periodic corpus, treating it as the coordinate reveals an open one. We report both throughout. In the target-aggregated diagnostic, a class is a set of spellings rather than a point on the line. After true fifths-circle, chromatic, set-valued line, spelling, frequency and tokenizer controls, a small held-out conditional contribution remains in some models; the corrected analysis does not license a domain-wide statement.

Methodological implications. Three instrument lessons carry beyond music. A categorical feature that occupies a contiguous block of the chosen cyclic ordering can fake the fundamental of the competing ordering; the cure is to project out or respell candidate features and to report mode isotropy, with the caveat that our two anisotropy diagnostics do not fully agree and that spectral sparsity does not guarantee separability (Fu et al., 2026). The last token of a multi-token concept name is not a neutral probe when token types differ, and neither is the span mean; the shared prompt-final position is. And predictor names are not definitions: a column called "circle" turned out to be a chromatic distance, and a merged class cannot be given an arithmetic line coordinate without saying what the merge means. Fold-local preprocessing and executable landmark tests are part of the scientific specification, not software hygiene.

Relation to the two theories. Karkada's construction is theory for embedding models and their LLM evidence is qualitative; Zhao et al.'s geometry is a statement about support patterns that are degenerate at this vocabulary size. Our results are empirical alignments between operators and observables, consistent with the qualitative idea that recovered geometry depends on which statistical object the computation is aligned with. We have not tested either factorization claim and do not manufacture a unification.

Limitations. We studied one unusually structured domain; nothing here shows that other families behave the same way. The behavioural task is constructed, not open-ended language use. The corpora are proxies even for OLMo, and the cross-corpus tests rest on fifteen paired source rows. The held-out test has fifteen source rows in all, and its positive cells lean on the rarest of them. The target-aggregated view changes the task, cannot score the enharmonic partner of a twin source, and its line controls necessarily summarize a set of spellings; v4 uses defensible extrema and endpoints, but no finite feature set exhausts tonal theory. The pointwise relabeling null conditions on the observed corpus; the document-cluster intervals quantify corpus sampling, are shifted toward zero by construction, and disagree between the percentile and pivotal forms; and nothing survives a correction over the full set of tests run. The KL scorer was implemented after the rank scorer had been seen. Correspondence is not causation. The checkpoint evidence is one run whose released model is a different run, and the corrected residual trajectory is weak. Hidden-state claims remain deliberately bounded.

Why the failed hypotheses are in the paper. The "circle-to-line transition", "predicting-position locus", "scaling trend" and "black/white-key feature" readings were killed by matched contexts, layer-selection nulls and respelling; lexical-code and frozen-transient explanations were killed by experiment; the training-data fingerprint was not established. The Phase-V headline was weakened twice more, first when review exposed a wrong circle predictor and an indefensible merged-line centroid, then when a row jackknife showed the corrected residual resting on one key. The auditable paper-v1 snapshot is retained; this version reports the corrected experiment rather than hiding the failure.

What to take away. Three habits transfer beyond music. Before reading a circle off a Fourier spectrum, respell or project out the categorical features that are contiguous on the ordering; a block is a harmonic. Before calling any statistic "the" geometry of a concept family, compute a second one built differently, and note which the behaviour follows; here the first-order and second-order statistics of one corpus disagree, and behaviour picks the second. And when a held-out gain is a mean over a handful of rows, look at the rows before believing the mean.

Reproducibility

All code, results files, figures and logs are in the public repository `github.com/VykosMolt/One-Concept-Multiple-Geometries` (tag `paper-v2.4`; `paper-v2` to `paper-v2.3` are the earlier corrected drafts); the immutable `paper-v1` tag preserves the pre-correction manuscript. Model weights, corpora, extracted hidden-state tensors, the multi-context activation files behind Result 3's principal-axis statement and the merged count matrices behind Result 1's helper factorization are excluded from the mirror; the mirror's `MIRROR_MANIFEST.md` lists each excluded file with its size and SHA-256 digest, and the scripts that regenerate them from the public models and corpora are included. The chronological logs state which analyses were fixed before or after data were seen; `V4_CORRECTION_REPORT.md` and `results/phase5/v3_v4_comparison.*` expose the correction and every inherited numerical change. Figures are regenerated from v4 artifacts by `paper/make_figs.py`. The v4 regressions/nulls used CPU only (0 GB accelerator memory); earlier model forward passes used one RTX 5070 Ti laptop GPU.

References

Emmanuel Amiot. *Music through Fourier space: Discrete Fourier transform in music theory*. Computational Music Science. Springer, 2016.

Saksham Bassi and Sharvi Tomar. Geometry of ordinal representations in language models. In *Mechanistic Interpretability Workshop, 43rd International Conference on Machine Learning*, 2026. arXiv:2607.04167.

Sasha Brenner, Thomas R. Knösche, and Nico Scherf. Predictive statistics shape emergent world representations of grid walkers. *arXiv preprint arXiv:2603.16689*, 2026.

Elaine Chew. *Towards a mathematical model of tonality*. PhD thesis, Massachusetts Institute of Technology, 2000.

Ching-Hua Chuan, Kat Agres, and Dorien Herremans. From context to concept: exploring semantic relationships in music with word2vec. *Neural Computing and Applications*, 32:1023–1036, 2020. doi: 10.1007/s00521-018-3923-1.

Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2405.14860.

Sheridan Feucht, Tal Haklay, Usha Bhalla, Daniel Wurgaft, Can Rager, Raphaël Sarfati, Jack Merullo, Thomas McGrath, Owen Lewis, Ekdeep Singh Lubana, Thomas Fel, and Atticus Geiger. Arithmetic in the wild: Llama uses base-10 addition to reason about cyclic concepts. *arXiv preprint arXiv:2605.01148*, 2026.

Deqing Fu, Tianyi Zhou, Mikhail Belkin, Vatsal Sharan, and Robin Jia. Convergent evolution: How different language models learn similar number representations. In *Conference on Language Modeling (COLM)*, 2026. arXiv:2604.20817.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

Wes Gurnee and Max Tegmark. Language models represent space and time. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.02207.

Zhimin Hu, Lanhao Niu, and Sashank Varma. Language models represent and transform concepts with shared geometry. *arXiv preprint arXiv:2607.04525*, 2026.

Cheng-Zhi Anna Huang, David Duvenaud, and Krzysztof Z. Gajos. ChordRipple: Recommending chords to help novice composers go beyond the ordinary. In *Proceedings of the 21st International Conference on Intelligent User Interfaces (IUI)*, pages 241–250, 2016. doi: 10.1145/2856767.2856792.

Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language models struggle to learn long-tail knowledge. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023. arXiv:2211.08411.

Dhruva Karkada, Daniel J. Korchinski, Andres Nava, Matthieu Wyart, and Yasaman Bahri. Symmetry in language statistics shapes the geometry of model representations. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.15029.

Carol L. Krumhansl and Edward J. Kessler. Tracing the dynamic changes in perceived tonal organization in a spatial representation of musical keys. *Psychological Review*, 89(4):334–368, 1982.

Omer Levy and Yoav Goldberg. Neural word embedding as implicit matrix factorization. In *Advances in Neural Information Processing Systems 27*, pages 2177–2185, 2014.

Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, et al. DataComp-LM: In search of the next generation of training sets for language models. *arXiv preprint arXiv:2406.11794*, 2024.

Raja Marjieh, Veniamin Veselovsky, Thomas L. Griffiths, and Ilia Sucholutsky. What is a number, that a large language model may know it? *arXiv preprint arXiv:2502.01540*, 2025.

Fabian C. Moss, Markus Neuwirth, and Martin Rohrmeier. The line of fifths and the co-evolution of tonal pitch-classes. *Journal of Mathematics and Music*, 17(2):173–197, 2023. doi: 10.1080/17459737.2022.2044927.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2301.05217.

OLMo Team, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, et al. 2 OLMo 2 Furious. *arXiv preprint arXiv:2501.00656*, 2025.

Lucas Prieto, Edward Stevinson, Melih Barsbey, Tolga Birdal, and Pedro A. M. Mediano. From data statistics to feature geometry: How correlations shape superposition. *arXiv preprint arXiv:2603.09972*, 2026.

Ian Quinn. General equal-tempered harmony (introduction and part I). *Perspectives of New Music*, 44(2):114–158, 2006.

Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Najla Sadek and Joseph Bakarji. The circle of fifths as latent geometry in Bach's Well-Tempered Clavier. In *Proceedings of Machine Learning Research*, volume 303, pages 1–13. PMLR, 2026. URL https://proceedings.mlr.press/v303/sadek26a.html. 1st International Workshop on Emerging AI Technologies for Music, Singapore.

Magnus Sahlgren. The distributional hypothesis. *Italian Journal of Linguistics*, 20(1):33–53, 2008.

Hinrich Schütze and Jan Pedersen. A vector model for syntagmatic and paradigmatic relatedness. In *Proceedings of the 9th Annual Conference of the UW Centre for the New OED and Text Research*, pages 104–113, 1993.

Adam Shai, Loren Amdahl-Culleton, Casper L. Christensen, Henry R. Bigelow, Fernando E. Rosas, Alexander B. Boyd, Eric A. Alt, Kyle J. Ray, and Paul M. Riechers. Transformers learn factored representations. *arXiv preprint arXiv:2602.02385*, 2026.

Simardeep Singh and Paras Chopra. Geometry of human perceptual domains emerges transiently in LLM representations. *arXiv preprint arXiv:2605.27970*, 2026.

David Temperley. The line of fifths. *Music Analysis*, 19(3):289–319, 2000.

Daniel Wurgaft, Can Rager, Matthew Kowal, Vasudev Shyam, Sheridan Feucht, Usha Bhalla, Tal Haklay, Eric Bigelow, Raphaël Sarfati, Thomas McGrath, Owen Lewis, Jack Merullo, Noah Goodman, Thomas Fel, Atticus Geiger, and Ekdeep Singh Lubana. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv preprint arXiv:2605.05115*, 2026.

Yize Zhao and Christos Thrampoulidis. Geometry of semantics in next-token prediction: How optimization implicitly organizes linguistic representations. *arXiv preprint arXiv:2505.08348*, 2025.

Yize Zhao, Tina Behnia, Vala Vakilian, and Christos Thrampoulidis. Implicit geometry of next-token prediction: From language sparsity patterns to model representations. *arXiv preprint arXiv:2408.15417*, 2024.

Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems 36*, 2023. arXiv:2306.17844.

APPENDIX A Audit trail: retracted claims and post-hoc decisions

Two independent model-assisted adversarial audit passes (a code and instrument verification; a scientific review), three manuscript audits, and the Opus review that triggered v4 were run; these are not human peer review and are not offered as a credential. Their summaries are retained in the repository, including `notes/review_opus5_v4.md` and `V4_CORRECTION_REPORT.md`. The table lists claims later withdrawn, corrections, and decisions taken after seeing data.

Claim or decision	Status	What happened
Key-name activations contain a circle of fifths (P_5 share 0.33, $z \approx 3.7$)	Retracted	Accidental-indicator projection: P_5 falls to the matched null; respelling shows the block is orthographic (§4).
"Projected P_5 is below the null"	Corrected	The projection-matched null is 0.120, not 2/11; observed 0.11–0.14 is *at* the null ($z = +0.4$).
"Projection removes 13.5% of a circle's energy"	Corrected	It removes 47.9% of the *energy* (13.5% of the share); the projection is not mild, the share-based conclusion stands.
Black/white-key semantic block	Retracted	Respelling C, E, F, B as B♯, F♭, E♯, C♭ moves the block with the glyph in all four models.
Fifths geometry scales with model size	Retracted	With max-over-layers nulls three of four points are at $p = 0.09$–0.21; only the 7B is significant.
The circle lives at the predicting position	Retracted	Matched contexts: concept token +0.33…+0.52 vs predicting token +0.38…+0.53; the difference was context, not position.
Circle mid-network, line at the output	Retracted	Significant in 1 of 12 model×context cells after correction.
Line in key-name geometry of Qwen/7B (six families)	Retracted	Carried by the accidental token being the last token of two-token keys; ≈ 0 under a span-mean probe.
Modulation context strengthens the line at the key token	Retracted	Same last-token artifact (+0.21 → −0.10 at span mean); prompt-final effects retained (small).
Separable circle/line/spelling subspaces (H5)	Unresolved	Cross-fitted subspaces overlap; variance shares sum above one.
Arbitrary output codes carry the line (Phase II)	Retracted	First pass listed the codebook in line order; with randomized headers a list-position heuristic dominates; inconclusive, superseded by Appendix E.
"The corpus contains no task-conditioned line"	Corrected	True under the convention that books spelling class as orthography; under the tonal-pitch-class convention the directional rows are line-like (§6).
Few-shot relation accuracies	Bounded	Raw scorer is length-biased (multi-token names under-selected 1.6–7×); reported as a lower bound with a length-normalized companion; a "calibrated" variant over-corrected and is kept only as a failed method.
Corpus P_5 share 0.50	Bounded	Depends on the PMI diagonal; the controlled partials are the evidence.
M^* "2.00 ± 0.01", "κ monotone"	Corrected	±0.03, median $\rho \approx 1560$; one wobble at $d' = 5$.
`partial()` returned 0.04 on constant input	Corrected	NaN guard added; no reported number affected.
Bootstrap SDs of corpus statistics	Corrected	Poisson on weighted counts ignored document clustering; replaced by a shard-cluster bootstrap (`scripts/corpus_cluster_boot.py`); the canonical-family $P(\text{circle} > \text{line})$ falls from 0.87 to 0.63, merged-family results unchanged.
Spelled outputs alone induce the line (H6)	Retracted	Phase III: converged behaviour follows the oracle regardless of code.
The line is a frozen transient of rare-alias learning	Retracted	Phase IV: power law in exposures; aligned code accelerates; unique-state control; C♭/B rows differ in the corpus.
Held-out rank scorer ΔCV	Diagnostic	≈ 0; insensitive to distributional re-weighting of 11–14 targets.
Held-out KL scorer	Corrected	Named in the Phase-V design, implemented only at the v3 rerun after the rank result had been seen; nulls and rich-theory check applied afterwards.
Within-row R^2 gain	Diagnostic	Not named in the design; added with the KL scorer and declared post-hoc.
Wrong circle predictor in paper-v1 (B1)	Corrected	The named circle column was chromatic cyclic distance. V4 uses $f(z) = 7z \bmod 12$ for the true fifths circle and retains chromatic distance separately; C–G and C–D♭ landmark tests prevent recurrence.
Merged-target line centroid (B2)	Corrected	Arithmetic means assigned artificial coordinates to distant enharmonic spellings. V4 treats a target class as a set and uses nearest/farthest distances and signed interval endpoints, never a centroid.

Claim or decision	Status	What happened
Global predictor standardization	Corrected	V4 fits tokenizer residuals and every predictor's mean/scale on the 14 training source rows inside each fold; the held-out response enters no preprocessing.
Process-dependent RNG seeds	Corrected	Python's salted `hash()` was replaced by SHA-256 derivation from master seed `20260830` and explicit cell/replicate labels; nulls and five-seed thinnings were regenerated.
Spelled-view held-out gain	Bounded	R^2 and KL scorers disagree in two of eight corrected modulation cells; no broad spelled-view claim is made.
Training-data fingerprint	Retracted	Eight of 72 corrected DiD cells are nominally significant, split four OLMo-favouring/four opposing; 33 contrasts are positive and 39 negative. No multiplicity correction is passed.
Reporting 4 of 9 DiD cells per corpus	Corrected	The first draft restricted the specificity test to harmonic/modulation × window/document; the restriction was post-hoc and is withdrawn; all cells are reported (Appendix F.2, Appendix C.3).
54-shard DCLM expansion	Diagnostic	Scanned after the first cross-corpus analyses; declared in the log.
"$p < .001$" from 300 relabelings	Corrected	The first held-out null used $B = 300$ and b/B (floor 0.0033); all Phase-V p-values use $B = 5{,}000/2{,}000$ and $(b+1)/(B+1)$. V4 values are reported only to their available resolution.
"Symmetric vs directional" framing	Corrected	A symmetrized conditional and a PMI built from the same 40-word counts show the contrast is conditional-row vs association, not directionality (§6); title, abstract and central claim revised.
Target-popularity robustness	Bounded	A training-row target prior preserves positive window gains in OLMo-1B, Gemma and OLMo-7B but does not rescue Qwen; document results remain mixed (Appendix F).
Figure F.1(c) showed only the favourable spelled-view scorer	Corrected	Both scorers now plotted.
Released OLMo-2-1B = stage-2 endpoint ("soup")	Corrected	Sampled-tensor relative distances show the ingredient endpoints are mutually close and their mean is no closer to the released weights; the released run's provenance is otherwise not reconstructed.
Respelling "moves the block with the glyph"	Bounded	The surviving block follows the glyph, but at a third to four fifths of its strength and confounded with a large drop in name frequency (round-5 review).
"Directionality contributes nothing"	Bounded	True here because the 40-word co-mention matrix is 98% symmetric; not a general separation of construction from direction (round-5 review).
Held-out residual in three of four models	Bounded	Row jackknife: C♭ major carries 66/44/94% of the OLMo-1B/Gemma/OLMo-7B gain; only Gemma's is distributed; percentile and pivotal bootstrap intervals disagree; nothing survives a 288-test correction (round-5 review).
"Circle or line adds ≤ 0.04 in every cell"	Corrected	True of layer means (max +0.03); 30 of 660 single layer-cells exceed it (max +0.10); the quoted R^2 range was not a computable summary and is restated (round-5 review).
Stated orthographic control family	Corrected	No analysis used the family as listed in §2; the per-analysis sets are now given in Appendix D.7 (round-5 review).
"Conditional-row operators keep twins apart"	Bounded	A document-cluster bootstrap of the conditional ECI includes the chance value ($[0.47, 0.67]$ directional); the operators do not identify twins, and a marginals-only null shows the construction does not manufacture the separation (round-5 review).
Ridge $\lambda = 1$ as regularization	Diagnostic	Effectively OLS on this design (eigenvalues 1.6–730); a λ sweep leaves the pattern unchanged; disclosed in D.6 (round-5 review).
Twin-difference vectors as fingerprint	Unresolved	Line-controlled alignment significant only in scattered cells and checkpoints.

APPENDIX B Fourier bookkeeping and nulls

Automorphisms. The units of $\mathbb{Z}_{12}$ are $\{1, 5, 7, 11\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$: chromatic up, fourths up, fifths up, chromatic down. If $H'[x] = H[7x \bmod 12]$ then $\hat{h}'_k = \hat{h}_{7k \bmod 12}$, the induced permutation on modes being $k \mapsto 0, 7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5$; hence $P_1 \leftrightarrow P_5$, while P_2, P_4, E_6 are fixed and P_3 maps to itself as a pair. The unit 5 induces the same swap and 11 is conjugation, so all four automorphic orderings give the same paired spectrum up to the single swap. Parseval, conjugate symmetry, the mode permutation and a pure fifths-fundamental test vector are unit-tested.

Karkada statistic and prediction. $M^*_{ij} = (P_{ij} - P_i P_j)/\frac{1}{2}(P_{ij} + P_i P_j) = 2(\rho - 1)/(\rho + 1)$ with $\rho = P_{ij}/P_i P_j$, bounded in $[-2, 2]$ and $\approx$ PMI for ρ near 1. Under translation symmetry the predicted mode energies are $E_k = |\lambda_k|$ for the circulant kernel's Fourier coefficients (relative amplitudes only; $|M^*|_S \neq |M^*_S|$ in general, and the block prediction can be bypassed by helper words).

Aliasing. For a feature f over $\mathbb{Z}_{12}$ let $F_k = \sum_x f(x)e^{-2\pi ikx/12}$. A contiguous block of width w has

$$|F_1|^2 \;\propto\; \frac{\sin^2(\pi w/12)}{\sin^2(\pi/12)},$$

which gives fundamental-pair shares of 18/37/55/70/80/83% for $w = 1, \ldots, 6$. The accidental indicator is the $w = 5$ block in fifths coordinates, hence 79.6% in $k = 5/7$ under semitone labelling.

Nulls. Under uniform relabeling $E[E_k] = \|H_c\|^2/11$ for every $k \geq 1$, so the null paired profile is $(2/11, \ldots, 2/11, 1/11)$; after projecting out a direction that carries 79.6% of its energy in P_5 the null P_5 share drops to 0.120 (the projection is applied inside every permutation). Partial correlations on 66 or 105 pairs carry free and block-preserving relabeling nulls; best-over-layers values carry a max-over-layers null; variance inflation of the control design is ≤ 1.4 (circle/line ≈ 2.5).

APPENDIX C — Full tables

C.1 Key-name geometry after controls (12-key design)

Model	glyph RSA	best-layer fifths	p (free / block)	final layer	white-key line, p
OLMo-2-1B	+0.86	+0.21 (L13)	0.16 / 0.21	−0.16	+0.41, 0.19
Gemma-2-2B	+0.84	+0.29 (L23)	0.10 / 0.09	−0.13	+0.61, 0.028
Qwen2.5-3B	+0.84	+0.23 (L14)	0.15 / 0.19	−0.16	+0.42, 0.24
OLMo-2-7B	+0.86	+0.34 (L30)	0.046 / 0.018	+0.09	+0.78, 0.001
Corpus PMI (canonical / merged)	+0.47	+0.62 / +0.66	$\leq$ 0.0005	—	+0.63, 0.016
Helper factorization ($d = 300$)	+0.57	+0.52	$\leq$ 0.0005	—	+0.45

Corpus and helper rows have no layers: their "best-layer fifths" entry is the single partial and their "glyph RSA" is the mutual RSA of the corpus block with the glyph geometry. The white-key column reports the layer-corrected p; exact-null p values are 0.037/0.003/0.025/<0.001 for the four models.

C.2 Held-out prediction, Wikipedia, target-aggregated view (modulation family)

Theory-only held-out KL per row and ΔKL from adding the corpus row; relabeling $p = (b+1)/(B+1)$ with $B = 5{,}000$ in parentheses. TP adds each target's mean log-probability over the 14 training source rows. Document-cluster bootstrap intervals are in Table C.8. Slash-separated entries are base/rich.

Model	Extraction	KL_0 b/r	ΔKL base (p)	ΔKL rich (p)	ΔKL TP b/r (p)
OLMo-2-1B	window	.0631/.0585	+.0041 (.0150)	+.0035 (.0340)	+.0043/.0039 (.0002/.0004)
OLMo-2-1B	document	.0631/.0585	+.0052 (.0076)	+.0039 (.0440)	+.0053/.0039 (.0002/.0002)
Gemma-2-2B	window	.0512/.0468	+.0053 (.0012)	+.0061 (.0004)	+.0037/.0045 (.0004/.0002)
Gemma-2-2B	document	.0512/.0468	+.0041 (.0118)	+.0031 (.0224)	+.0025/.0019 (.0004/.0010)
Qwen2.5-3B	window	.0452/.0420	−.0001 (.3339)	−.0002 (.4023)	−.0004/ − .0003 (.6739/.6079)
Qwen2.5-3B	document	.0452/.0420	+.0000 (.3033)	−.0003 (.5535)	−.0002/ − .0004 (.3479/.7083)
OLMo-2-7B	window	.0495/.0455	+.0038 (.0030)	+.0033 (.0026)	+.0033/.0030 (.0006/.0010)
OLMo-2-7B	document	.0495/.0455	+.0007 (.2346)	−.0003 (.6283)	+.0013/.0004 (.0082/.0446)

C.3 Cross-corpus held-out gain and difference-in-differences (nats per row)

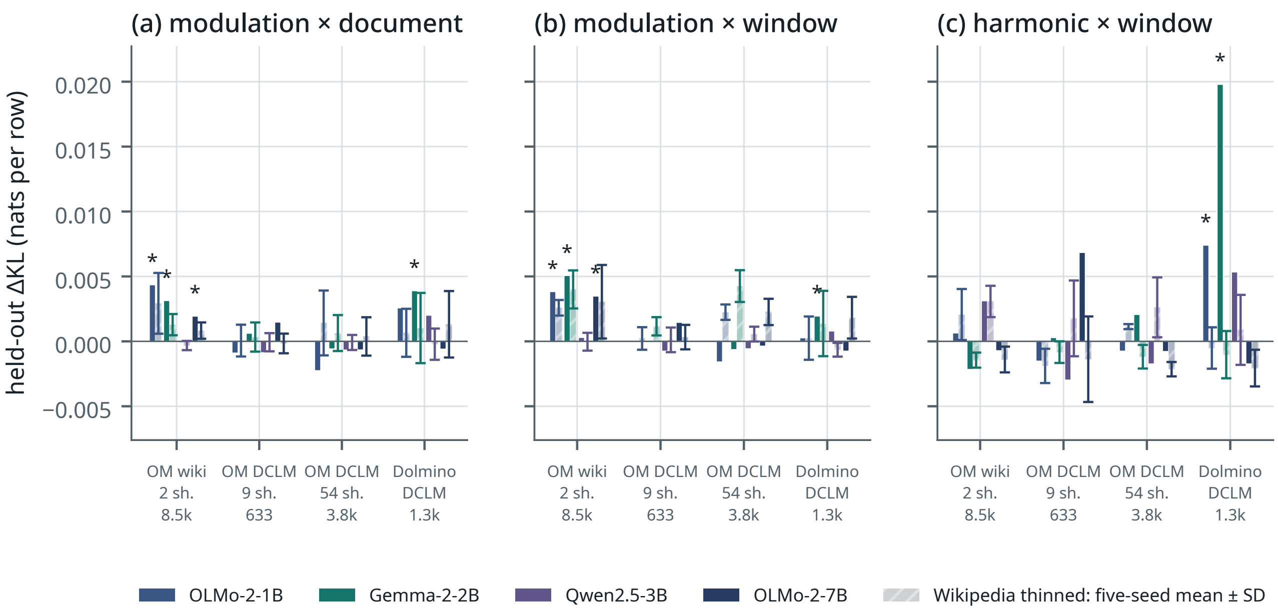

Figure C.1. Corrected cross-corpus held-out gain (target-aggregated view). Panels show modulation×document, modulation×window and harmonic×window. Solid: the direct corpus estimate; hatched bar and whisker: mean ± sample SD of five independently thinned Wikipedia baselines. Pair counts in the tick labels are window-extraction pair masses and apply to all three panels. Gains vary in sign and significance across corpora, operators and models; the figure does not support universality or corpus specificity.

DiD = (OLMo models − Gemma/Qwen) × (corpus − five-seed mean of size-matched Wikipedia); two-sided Wilcoxon over 15 source rows. The full target-aggregated set follows; exact direct, baseline-seed and specificity values for both views are in `results/phase5/crosscorpus_compare_v4_aggregated.json` and `crosscorpus_compare_v4_spelled.json`.

Corpus / family	window	cue	document
OLMo-Mix wiki, harmonic	−0.0000 (.524)	+0.0015 (.229)	+0.0016 (.720)
OLMo-Mix wiki, chord	−0.0017 (.762)	−0.0032 (.524)	+0.0015 (.679)
OLMo-Mix wiki, modulation	+0.0001 (.934)	+0.0019 (.454)	+0.0002 (.489)
OLMo-Mix DCLM (9 sh.), harmonic	+0.0061 (.389)	+0.0039 (.454)	+0.0040 (.208)
OLMo-Mix DCLM (9 sh.), chord	+0.0001 (.679)	+0.0036 (.599)	−0.0009 (.561)
OLMo-Mix DCLM (9 sh.), modulation	+0.0014 (.330)	−0.0005 (.489)	+0.0006 (.252)
OLMo-Mix DCLM (54 sh.), harmonic	+0.0003 (.934)	−0.0022 (.454)	+0.0032 (.454)
OLMo-Mix DCLM (54 sh.), chord	−0.0024 (.934)	−0.0008 (.934)	−0.0012 (.252)
OLMo-Mix DCLM (54 sh.), modulation	−0.0002 (.978)	+0.0006 (.208)	−0.0015 (.454)
Dolmino DCLM, harmonic	−0.0085 (.135)	+0.0006 (.934)	−0.0073 (.188)
Dolmino DCLM, chord	+0.0047 (.303)	+0.0025 (.639)	+0.0091 (.018)
Dolmino DCLM, modulation	−0.0023 (.073)	+0.0009 (.121)	−0.0025 (.022)

Significant spelled-view DiD cells: 9-shard OLMo-Mix DCLM modulation×window +0.0072 (p = .0054); Dolmino chord×window +0.0077 (.018), chord×cue +0.0037 (.026), and modulation×document −0.0072 (.035); 54-shard OLMo-Mix DCLM chord×document −0.0106 (.0015) and modulation×document −0.0141 (.0006). Thus the six split three positive/three negative; together with the target-aggregated table, the eight significant cells split four/four. The 54-shard sample is a declared post-hoc expansion.

C.4 Synthetic factorial, final checkpoint (means over five seeds)

Condition	KL to oracle	circle | line	line | circle (oracle)	twin-source JS	twin-target asym.
CIRCLE × aligned	0.001	0.976	0.436 (0.439)	0.000	0.101
CIRCLE × permuted	0.001	0.979	0.422 (0.439)	0.000	0.088
LINE × aligned	0.000	0.000	0.995 (0.995)	0.564	3.22
LINE × permuted	0.001	−0.008	0.995 (0.995)	0.564	3.23

The circle | line column is the code-controlled partial; the line | circle column is the uncontrolled partial, the one comparable across code conditions (Appendix E); the code-controlled line | circle values are 0.113 / 0.421 / 0.787 / 0.995.

C.5 OLMo-2-1B checkpoints (modulation family)

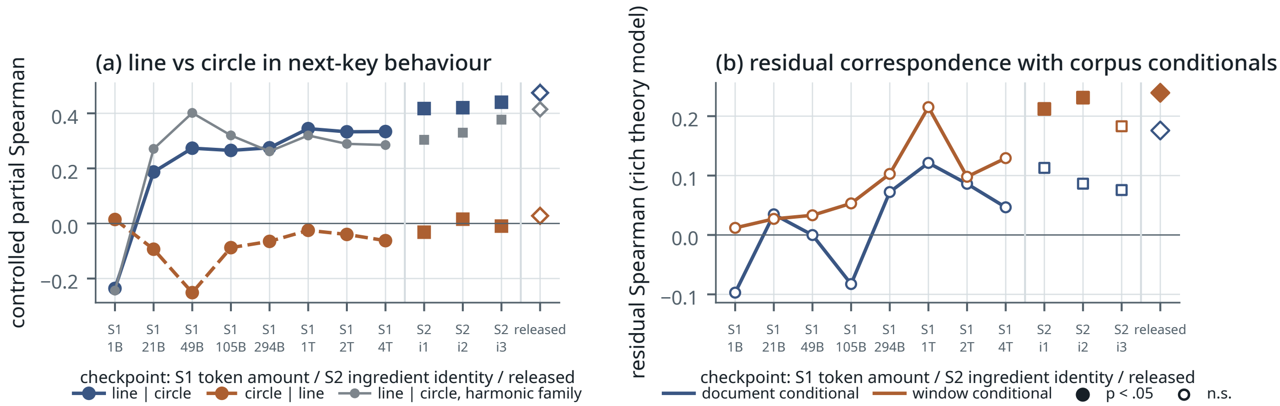

Figure C.2. OLMo-2-1B checkpoints, modulation family. (a) The open-line coordinate appears by 21B tokens and rises in two steps (blue; grey: the harmonic-relation family); the circle partial (orange) has one −0.25 excursion. (b) Residual correspondence under the corrected rich baseline is weak and noisy: no stage-1 point is significant, two of three stage-2 window points are, and no document point is. Filled markers denote $p < .05$ ($B = 2{,}000$). Squares: stage-2 ingredients; diamond: the released model, a different training run.

Provenance of the released 1B model. A 22-tensor check finds the three ingredient endpoints mutually close (median relative L2 distance 0.096), while each endpoint and their arithmetic mean are about 1.12 from the released `OLMo-2-0425-1B`. This supports treating the released model as neither the branch endpoint nor a simple three-way arithmetic soup; it does not reconstruct the released run's provenance. The released model is plotted separately (line | circle +0.47, corrected window residual +0.24), which complicates rather than completes the acquisition story.

Checkpoint (tokens)	line | circle	circle | line	residual, window (p)	residual, document (p)
stage 1, 1B	−0.24	+0.01	+0.01 (.509)	−0.10 (.599)
stage 1, 21B	+0.19	−0.09	+0.03 (.453)	+0.03 (.449)
stage 1, 49B	+0.27	−0.25	+0.03 (.460)	−0.00 (.562)
stage 1, 105B	+0.27	−0.09	+0.05 (.339)	−0.08 (.669)
stage 1, 294B	+0.27	−0.07	+0.10 (.319)	+0.07 (.389)
stage 1, 1007B	+0.34	−0.02	+0.22 (.083)	+0.12 (.239)
stage 1, 1993B	+0.33	−0.04	+0.10 (.214)	+0.09 (.290)
stage 1, 4001B	+0.33	−0.06	+0.13 (.245)	+0.05 (.434)
stage 2, ingredient 1 (51B)	+0.42	−0.03	+0.21 (.023)	+0.11 (.199)
stage 2, ingredient 2 (51B)	+0.42	+0.02	+0.23 (.013)	+0.09 (.238)
stage 2, ingredient 3 (51B)	+0.44	−0.01	+0.18 (.052)	+0.08 (.289)
released model (other run)	+0.47	+0.03	+0.24 (.018)	+0.18 (.138)

C.6 Scorer robustness of Result 3 (line | circle and circle | line, family means over four templates)

Total sequence log-probability scorer vs the pre-registered length-normalized scorer; "sig" = templates with line | circle significant under the glyph-preserving relabeling null ($B = 1{,}000$, b/B); ECI = enharmonic collapse index of the behavioural matrix (total / length-normalized).

Model	Family	total line | circle	total circle | line	sig	norm. line | circle	norm. circle | line	sig	ECI
OLMo-2-1B	B enharmonic	+0.30	+0.07	4/4	+0.28	+0.08	3/4	0.70/0.58
OLMo-2-1B	C harmonic	+0.33	−0.03	4/4	+0.21	−0.00	4/4	0.86/0.69
OLMo-2-1B	D chord	+0.27	−0.10	4/4	+0.19	−0.06	2/4	0.84/0.59
OLMo-2-1B	E modulation	+0.44	+0.02	4/4	+0.37	+0.01	4/4	0.88/0.81
Gemma-2-2B	B enharmonic	+0.50	−0.14	4/4	+0.45	−0.07	4/4	0.83/0.75
Gemma-2-2B	C harmonic	+0.32	−0.00	4/4	+0.24	+0.03	4/4	0.83/0.73
Gemma-2-2B	D chord	+0.36	−0.14	4/4	+0.26	−0.03	4/4	0.90/0.72
Gemma-2-2B	E modulation	+0.54	−0.07	4/4	+0.43	+0.03	4/4	0.90/0.85

Model	Family	total line | circle	total circle | line	sig	norm. line | circle	norm. circle | line	sig	ECI
Qwen2.5-3B	B enharmonic	+0.31	+0.11	4/4	+0.31	+0.07	4/4	0.51/0.38
Qwen2.5-3B	C harmonic	+0.42	+0.11	4/4	+0.27	+0.09	4/4	0.73/0.52
Qwen2.5-3B	D chord	+0.40	−0.01	4/4	+0.23	+0.06	3/4	0.78/0.57
Qwen2.5-3B	E modulation	+0.52	+0.07	4/4	+0.50	+0.00	4/4	0.88/0.79
OLMo-2-7B	B enharmonic	+0.14	+0.28	3/4	+0.19	+0.25	3/4	0.11/0.05
OLMo-2-7B	C harmonic	+0.38	+0.16	4/4	+0.38	+0.09	4/4	0.67/0.43
OLMo-2-7B	D chord	+0.50	+0.13	4/4	+0.38	+0.16	3/4	0.74/0.53
OLMo-2-7B	E modulation	+0.56	+0.13	4/4	+0.45	+0.12	4/4	0.81/0.68

C.7 Template robustness of the corrected held-out gain

Target-aggregated view, modulation family, base v4 ΔKL (nats per row) for OLMo-1B / Gemma / Qwen / OLMo-7B when the behavioural matrix is built from a single template or the leave-one-template-out mean; $^{\mathrm{ns}}$ marks relabeling $p \geq .05$ ($B = 2{,}000$). Single templates are significant in 9/32 model×extraction cells and leave-one-out aggregates in 19/32; Qwen is null in every aggregate. Templates follow Appendix D.4.

Templates	window conditional	document conditional
template 1 only	$+.005927/+.004077/-.000028^{\mathrm{ns}}/+.002964$	$+.005418/+.003622^{\mathrm{ns}}/+.000002^{\mathrm{ns}}/-.001356^{\mathrm{ns}}$
template 2 only	$+.007036^{\mathrm{ns}}/+.004024/+.000290^{\mathrm{ns}}/+.002864^{\mathrm{ns}}$	$+.006629^{\mathrm{ns}}/+.004657/-.000403^{\mathrm{ns}}/+.001842^{\mathrm{ns}}$
template 3 only	$+.000199^{\mathrm{ns}}/+.009538/-.001198^{\mathrm{ns}}/+.000284^{\mathrm{ns}}$	$+.002903^{\mathrm{ns}}/+.003544^{\mathrm{ns}}/+.000102^{\mathrm{ns}}/-.000897^{\mathrm{ns}}$
template 4 only	$+.002787^{\mathrm{ns}}/+.003387/-.000616^{\mathrm{ns}}/+.009899$	$+.002018^{\mathrm{ns}}/+.002444^{\mathrm{ns}}/-.000054^{\mathrm{ns}}/+.002778^{\mathrm{ns}}$
all but 1	$+.002839^{\mathrm{ns}}/+.005577/-.000263^{\mathrm{ns}}/+.003609$	$+.003810/+.003871/-.000094^{\mathrm{ns}}/+.001440^{\mathrm{ns}}$
all but 2	$+.003263/+.005757/-.000417^{\mathrm{ns}}/+.004036$	$+.004839/+.003871/+.000107^{\mathrm{ns}}/+.000270^{\mathrm{ns}}$
all but 3	$+.005656/+.003947/-.000030^{\mathrm{ns}}/+.005361$	$+.005440/+.003686/-.000029^{\mathrm{ns}}/+.000771^{\mathrm{ns}}$
all but 4	$+.004484/+.006011/+.000092^{\mathrm{ns}}/+.002242$	$+.006358/+.004619/-.000014^{\mathrm{ns}}/+.000218^{\mathrm{ns}}$

C.8 Row jackknife and interval types for the held-out gain (target-aggregated view, modulation family)

ΔKL (nats per row) with and without the C♭ source row, and the two 95% document-cluster bootstrap intervals from the same 300 draws. Slash-separated entries are base/rich. Source: `results/phase5/row_jackknife_v4.txt`.

Model	Extraction	ΔKL b/r	without C♭	percentile CI b/r	pivotal CI b/r
OLMo-2-1B	window	.0041/.0035	.0021/.0013	[+.0002, +.0056]/[-.0009, +.0054]	[+.0026, +.0079]/[+.0015, +.0078]
OLMo-2-1B	document	.0052/.0039	.0035/.0019	[-.0027, +.0091]/[-.0018, +.0078]	[+.0013, +.0130]/[-.0000, +.0096]
Gemma-2-2B	window	.0053/.0061	.0038/.0037	[+.0015, +.0068]/[+.0016, +.0071]	[+.0039, +.0091]/[+.0050, +.0105]
Gemma-2-2B	document	.0041/.0031	.0020/.0009	[-.0009, +.0070]/[-.0015, +.0057]	[+.0012, +.0090]/[+.0006, +.0078]
Qwen2.5-3B	window	-.0001/-.0002	-.0002/-.0003	[-.0010, +.0005]/[-.0010, +.0003]	[-.0008, +.0007]/[-.0007, +.0007]
Qwen2.5-3B	document	.0000/-.0003	.0000/-.0003	[-.0012, +.0016]/[-.0014, +.0006]	[-.0016, +.0013]/[-.0012, +.0008]
OLMo-2-7B	window	.0038/.0033	.0008/.0002	[+.0002, +.0050]/[-.0006, +.0046]	[+.0026, +.0074]/[+.0021, +.0073]
OLMo-2-7B	document	.0007/-.0003	-.0006/-.0013	[-.0021, +.0028]/[-.0024, +.0018]	[-.0014, +.0035]/[-.0024, +.0018]

APPENDIX D Methods in full

D.1 Keys, spellings and aliases

The 12-key design uses the conventional spellings C, D♭, D, E♭, E, F, F♯, G, A♭, A, B♭, B (semitone $x = 0, \ldots, 11$; fifths position $7x$ mod 12; signed accidental count $s = p$ for $p \leq 6$, $p - 12$ otherwise). The 15-key design uses the standard major-key spellings on the signed line $s = -7, \ldots, +7$: C♭, G♭, D♭, A♭, E♭, B♭, F, C, G, D, A, E, B, F♯, C♯; enharmonic pairs (C♭, B), (G♭, F♯), (D♭, C♯). In text, symbol (C#, Db), word (C-sharp, D flat, C sharp) and Unicode (C♯, D♭) forms are all mapped to the symbol spelling. In the merged 12-key family the two spellings of each pitch class are pooled (B♯/C, D♭/C♯, E♭/D♯, F♭/E, E♯/F, G♭/F♯, A♭/G♯, B♭/A♯, C♭/B) before counting.

D.2 Corpus matching and symmetric statistics

Corpus: the `wikimedia/wikipedia 20231101.en` parquet release (41 shards, 6.41M documents, ≈ 3.1B whitespace-delimited words). A key mention is a match of the pattern `[A-G](accidental)?□(major|minor)`, not preceded by an alphanumeric character or hyphen and not followed by a letter; matches at sentence-initial position (preceded, ignoring spaces, by a sentence-final or opening character, or at document start) are excluded from unigram and pair counts because "A major" is ambiguous there. Symmetric statistics follow Karkada et al. (2026): for every ordered token pair within a window of $L = 16$ words the pair count is incremented by $f(d) = 17 - d$ with d the word distance; Z is the f-weighted number of (position, offset) pairs and N the number of word tokens; $\rho_{ij} = (C_{ij}/Z)/((n_i/N)(n_j/N))$; $M^* = 2(\rho - 1)/(\rho + 1)$ and $\mathrm{PMI} = \log \rho$, with zero-count cells floored one unit below the minimum observed PMI. Group tallies (all documents, documents mentioning "circle of fifths"/"perfect fifth", and the rest) are kept separately.

D.3 Conditional (ordered key-mention) statistics

For each document containing " major", the major-key mentions are listed in order with their character offsets (15 spellings only). Family A: for each ordered pair of mentions (a, b) with b after a, if the segment between the end of a and the start of b contains at most 40 whitespace-delimited words, $N^{A}_{k(a),k(b)}$ is incremented by one (all such pairs, including repeated mentions; the scan stops at the first later mention beyond 40 words). Family B increments N^{B_c} for the same pairs when the intervening segment contains a cue from class c (modulation, chord, key-signature, enharmonic, relation; `phase5/scan_conditional.py` lists the cue strings); B_any is the union. Family C matches four relational regular expressions. Family D increments $N^{D}_{k,k'}$ once per document for every pair of keys whose first mentions occur in that order. The corpus predictor is $\log C_{ij} = \log \frac{N_{ij}+0.5}{\sum_{j'}(N_{ij'}+0.5)}$ over the 15 target spellings; in the target-aggregated view the columns are merged by log-sum-exp over each pitch class. Per-document count matrices are stored for the cluster bootstrap.

D.4 Behavioural scoring

For a template t of family f and source key x_i, the prompt is t with {x} replaced by the symbol spelling of x_i (e.g. "The piece modulates from D♭ major to"). Each of the 15 candidates is the string "␣x_j␣major" appended to the prompt and tokenized without special tokens; the model's log-softmax over the vocabulary is evaluated once on the concatenated sequence (teacher forcing) and the candidate score is the sum of the log-probabilities of the candidate's tokens at their positions. The *total* scorer uses this sum, the *length-normalized* scorer divides it by the candidate's token count, and the *enharmonic-merged behavioural scorer* log-sum-exps the two spellings of each twin before normalizing (applied to families B, C, D only). Scores are log-softmax-normalized over the 15 candidates within a row; the family matrix used in Appendix F is the mean over the family's four templates of the per-template row-normalized log-probabilities, renormalized. Precision: OLMo-2-1B and its checkpoints in fp32; Gemma-2-2B, Qwen2.5-3B and OLMo-2-7B in bf16 (a bf16/fp32 check on OLMo-2-1B changed Fourier shares by ≤ 0.002). Prompt templates:

Family	Templates ({x} = source key)
A spelling	The key signature of {x} major has · In {x} major, the accidentals written in the key signature are · Spelled correctly, the scale of {x} major uses the notes · The number of sharps or flats in the key signature of {x} major is
B enharmonic	On the piano, the tonic of {x} major is the same key as the tonic of · In equal temperament, {x} major sounds identical to · The enharmonic equivalent of {x} major is · Played on a keyboard, a piece in {x} major uses the same keys as a piece in
C harmonic	The dominant key of {x} major is · The subdominant key of {x} major is · A key closely related to {x} major is · The relative minor of {x} major shares its key signature with
D chord	In the key of {x} major, the tonic chord is usually followed by the chord of · A typical chord progression in {x} major moves from the tonic chord to the chord of · In {x} major, the perfect cadence resolves from the dominant chord to the chord of · Improvising over a song in {x} major, the guitarist played the chord of
E modulation	The piece modulates from {x} major to · The first movement is in {x} major and the second movement is in · The song, originally in {x} major, was later transposed to · The first song on the album is in {x} major. The second song is in
F generic	She said the piece was in {x} major and everyone · The recording of the sonata in {x} major was released in · Most listeners preferred the version in {x} major because · The manuscript of the symphony in {x} major is kept in

D.5 Models and hidden-state positions

`allenai/OLMo-2-0425-1B` (revision `main` as of 2026-08-28; checkpoint branches `stage1-step300-tokens1B`, `-step10000-tokens21B`, `-step23100-tokens49B`, `-step50000-tokens105B`, `-step140000-tokens294B`, `-step480000-tokens1007B`, `-step95000`

`0-tokens1993B`, `-step1907359-tokens4001B`, `stage2-ingredient{1,2,3}-step23852-tokens51B`); `unsloth/gemma-2-2b` (a bf16 safetensors mirror of `google/gemma-2-2b`, used because the original repository is gated; no fidelity check against the gated weights was run); `Qwen/Qwen2.5-3B`; `allenai/OLMo-2-1124-7B`. Hidden states are taken at every layer at three positions: the last token of the key name (the accidental token for two-token spellings), the mean over the key-name span, and the prompt-final token; token counts of each spelling under each tokenizer enter the theory features. Layer bands for context contrasts are the middle half of the network, fixed before the results.

D.6 The held-out regression of Appendix F

Response and corpus predictor. $y_{ij} = \log Q_{ij}$ is the family-mean row-normalized log-probability of target j given source i, over all ordered off-diagonal pairs (210 in the spelled view; 165 in the target-aggregated view, where j ranges over the 11 pitch classes other than the source's own). The single corpus column is $c_{ij} = \log C_{ij}$ from D.3.

Corrected v4 theory features. Let z be semitone pitch class, $f(z) = 7z \bmod 12$ its fifth-coordinate class, and $s \in [-7, 7]$ the signed open-line coordinate. The spelled view contains the true circular-fifths distance $d_5(i, j) = \min(|f(z_i) - f(z_j)|, 12 - |f(z_i) - f(z_j)|)$; chromatic cyclic semitone distance as a distinct predictor; $|s_j - s_i|$ and $s_j - s_i$; same glyph class; separate source/target flat and sharp indicators; same root; alphabet distance; unsigned key-signature-size difference; and source/target log frequency. Tokenizer controls are residuals from token count regressed on an intercept plus flat and sharp indicators using only the fold's training source keys, entered for source and target.

In the target-aggregated view, a target class T is explicitly a *set* of the available enharmonic spellings. It retains d_5 and the separate chromatic distance, then represents the open line by $\min_{t \in T} |s_t - s_i|$, $\max_{t \in T} |s_t - s_i|$, and the signed endpoints $\min_{t \in T}(s_t - s_i)$ and $\max_{t \in T}(s_t - s_i)$. Thus each term has a nearest, farthest or interval-endpoint interpretation for the log-sum-exp target class; no artificial class centroid is introduced. The remaining class features are minimum alphabet distance; minimum and maximum signature-size difference and edit distance; source flat/sharp; whether T contains a flat or sharp spelling; any matching glyph or root; source log frequency and log summed target-class frequency; and the source tokenizer residual plus the mean residual across T. The rich design adds d_5^2, the relevant line-distance square (spelled absolute distance; aggregated farthest distance), d_5 times the spelled/nearest class line distance, chromatic distance times that line distance, and $\cos(2\pi d_5/12)$ and $\cos(4\pi d_5/12)$. The third cosine was removed because it is exactly dependent in this design. The target-prior robustness control adds, for each held-out row, each target's mean $y_{\cdot j}$ over the 14 training rows only.

Fold-local preprocessing and estimator. For every held-out source row, tokenizer coefficients are fitted on the 14 training source keys. Every continuous theory column, the corpus column, and the target prior is then centred and scaled using only training-pair observations; constant training columns receive scale 1. The training statistics are applied unchanged to the held-out predictors. Nothing is standardized globally, and the held-out response never enters feature construction, scaling or the target prior. Ridge regression uses an unpenalized intercept and fixed $\lambda = 1$ on the standardized columns. With about 154 training pairs this penalty is nearly inert: the eigenvalues of $X^\top X$ for the base theory-plus-corpus design run from 1.6 to 730, so the corpus coefficient is shrunk by well under one percent and the estimator is effectively ordinary least squares; the rich design has one near-collinear direction (smallest eigenvalue 0.006) that the penalty does suppress, which is why rich and base results differ little. A sweep over $\lambda \in \{0.01, 0.1, 1, 10, 100\}$ leaves every flagship cell's sign and significance pattern unchanged from $\lambda = 0.01$ to 1 and inflates the gains at $\lambda \geq 10$, where the over-shrunk theory model leaves more for the corpus column to explain; $\lambda = 1$ is therefore at the conservative end (`results/phase5/ridge_lambda_sensitivity_v4.txt`). For each source i, models are fitted on the other 14 rows, predicted log-scores are log-softmax-normalized over the held-out targets, and $\mathrm{KL}(q_i \,\|\, \hat{q}_i)$ is computed on the same target set (10^{-12} inside the log). ΔKL is mean KL(theory) minus KL(theory + corpus); within-row R^2 and ΔCV are retained diagnostics. Cross-corpus and target-prior runs call this same fold implementation. Four base columns (source flat, source sharp, source log frequency and the source tokenizer residual) are constant within a source row and therefore cannot change a log-softmax-normalized held-out prediction; the frequency and tokenizer controls act through their target-side columns. Because a source's own pitch class is excluded from its targets, the enharmonic partner of a twin source is never scored in the aggregated view. Checkpoint residual correspondence is explicitly a non-held-out, full-pair diagnostic with rich features and source effects, and is not described as cross-validated prediction.

Nulls, bootstrap and randomness. The corpus count matrix's rows and columns are jointly relabelled, the corpus predictor and its fold-local scaler are rebuilt for every relabelling, and $p = (b + 1)/(B + 1)$ for b null statistics at least as large as observed. $B = 5{,}000$ for Wikipedia, 2,000 for cross-corpus, single-template, leave-one-template-out and checkpoint runs, 500 for Phase-II geometry and 1,000 for Phase-II behaviour. The v4 document-cluster bootstrap resamples the 9,168 key-mentioning documents with replacement and rebuilds count matrices, source frequency features and raw theory features in each of 300 draws; it was run for the modulation and harmonic-relation families (window and document extractions). Table C.2 reports the naive percentile interval $[q_{.025}, q_{.975}]$ of the replicates; Table C.8 adds the pivotal interval $[2\hat{\theta} - q_{.975},\ 2\hat{\theta} - q_{.025}]$ and the leave-one-source-row-out decomposition (`results/phase5/row_jackknife_v4.txt`).

Phase-I uses the shard-cluster bootstrap; Phase-II nulls use b/B and one permutation bank shared across cells. All v4 null and thinning streams use master seed `20260830`; per-cell seeds are the first eight bytes of SHA-256 over canonical JSON containing the master seed and explicit corpus, family, extraction, model, view and replicate labels. Python's salted `hash()` is never used. Residual correspondence regresses y and c on the same corrected feature design over all pairs and reports the Spearman correlation of the residuals under the same relabelling family.

D.7 Control sets by analysis

The partial correlations of §3–§6 do not all use the same orthographic controls. Result 1 (15-key PMI, `phase2/corpus15.py`) controls for glyph class, edit distance, same letter and commonness; Result 3 (behaviour, `phase2/behavior_fit.py`) and Result 4 (operators, `phase5/operators.py`) add alphabet distance; the key-name geometry and context contrasts of §4 (`phase2/geometry_fit.py`, `context_contrast.py`) add alphabet distance and token count. The has-accidental and key-signature-size features defined in `phase2/keys15.py` are not used as controls in Phases I–II (they enter the Phase-V regression of D.6). Adding both to the Phase-II behavioural partial raises line | circle in every model × family (modulation $+0.44 \rightarrow +0.59$, $+0.54 \rightarrow +0.58$, $+0.52 \rightarrow +0.53$, $+0.56 \rightarrow +0.64$), so the omission is conservative for Result 3. The two values quoted for the 15-key PMI line partial (+0.38 in §3, +0.39 in Result 4) differ by this control set.

APPENDIX E Two artifact explanations, tested causally

Two cheap explanations of the natural line remained after §5: that spelled output tokens alone could bend prediction away from a periodic statistic, and that the models are frozen part-way through learning rare enharmonic equivalences. We tested both with small transformers trained from scratch (4 layers, $d = 128$) on synthetic data with no music vocabulary: 15 source states $n = -7.. + 7$, latent classes $z = n \bmod 12$ (three duplicated labels), a shared query token whose residual is the predictive-state probe, and eight-symbol output codewords.

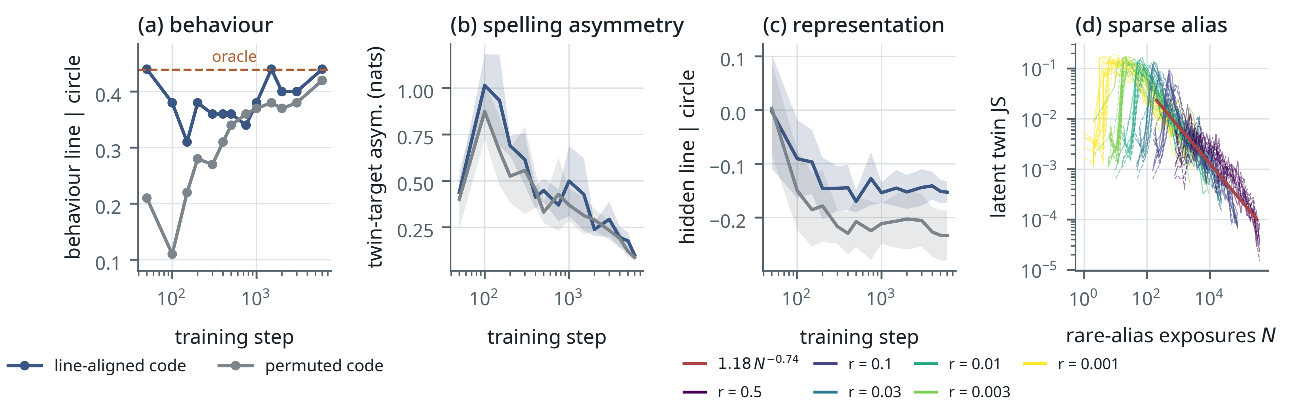

Figure E.1. Synthetic controls. (a–c) Periodic (CIRCLE) law, line-aligned vs permuted output code, five paired seeds: behaviour (line | circle without code control, seed means) reaches the oracle's own value faster under the aligned code and ends at the same value (a); the spelling asymmetry between twin targets is transiently larger and vanishes (b, mean ± sd); the predictive-state representation keeps a small persistent *relative* shift toward the line under the aligned code, both partials remaining negative and below their relabeling nulls (c, last layer, mean ± sd; the headline +0.097 contrast is at the mid layer). (d) Sparse source aliases: latent twin divergence collapses onto one power law in cumulative rare-alias exposures across rarity levels r (solid: aligned code; dashed: permuted).

Result 5 — output-code geometry: transient in behaviour, small and persistent in representation

We trained a 2 × 2 factorial of small transformers: a periodic (CIRCLE) or open (LINE) target law, crossed with an output code that is either aligned with the line or randomly permuted; five paired seeds; entropy-matched oracles (2.137 nats). All 20 runs reach KL ≤ 0.001 to their oracle. Under the periodic law, converged behaviour is periodic in both code conditions (circle | line 0.98; line | circle 0.436 vs 0.422 against the oracle's own 0.439; twin-target asymmetry +0.013, about 1%, in 5/5 seeds, paired t-test $p = .044$ but two-sided sign test $p = .0625$). Early in training the aligned code is closer to the oracle (KL 0.078 vs 0.276 at step 100) and twin asymmetry is larger through step ≈ 1500; every behavioural difference is gone by step 500–2000, and a dose-response over five alignment levels (15 runs, so five distinct levels) gives Spearman +0.85 for the early behavioural line and −0.68 for early KL. The predictive-state representation keeps a +0.08…+0.10 shift toward the line (5/5 seeds, paired t-test $p = .001$ at the mid layer; sign-test resolution $p = .0625$). That is a paired *difference*: the absolute partial stays negative in both conditions (-0.15 vs -0.23 at the last layer) and below each run's relabeling-null 95th percentile (+0.17…+0.21), so neither condition has a significant line; the code also imprints a quadratic bow ($|\rho(\mathrm{PC}_2, (n-\bar{n})^2)| = 0.85$ vs 0.39). Positive control: an arbitrary code does not hide a true line (0.995 recovered), though its bent open embedding registers as a spurious circle | line partial (+0.32), another instrument caution. Because the target law is stipulated, this experiment bounds a mechanism in a toy world; it says nothing directly about the natural line.

Result 6 — sparse aliases: rarity, not symmetry

We varied how rare a source alias is, $r \in \{0.5, \ldots, 0.001\}$, with latent-class mass and target law fixed (Wikipedia's own rare-spelling shares are C♭ within C♭/B = 0.024, C♯ within C♯/D♭ = 0.22 and G♭ within G♭/F♯ = 0.41); 90 runs. Global KL stays at the floor (0.001–0.007) while rare rows carry several times the error of common rows at $r \leq 0.01$ (3–24× across runs; about 1× at $r = 0.5$), so aggregate metrics do hide an unresolved alias, but the residual shrinks steadily. Pooling 1,254 checkpoints, latent twin JS $\approx 1.18\,N^{-0.74}$ in cumulative rare exposures N (partial Spearman with exposure given step -0.84; with step given exposure -0.19), and exposure-matched medians coincide across r. The aligned code *lowers* the error at every sparse r (significantly at $r = .01$, $p = .02$, and $r = .001$, $p = .035$; same sign but n.s. at $r = .003$, $p = .16$; fitted exponents -0.72 and -0.76, prefactors 0.94 and 1.5). A unique state with no twin is learned at least as well as a rare alias at matched exposure (row KL 0.0064 at 2.6k exposures for the unique state against 0.0105 at 2.3k for the rare alias of the $r = .003$ arm; 3–5 seeds, no uncertainty attached), so nothing transfers the twin's row to the rare alias. The representational twin distance collapses after 222–2,425 rare exposures whereas behavioural equalization needs ≈ 0.8k–7.7k, the representational threshold coming first in every run where both are crossed except one. This is the ordinary long-tail regime (Kandpal et al., 2023), not a symmetry-specific bottleneck. Separately, the natural premise fails: Wikipedia's next-key rows for C♭ and B are not the same predictive state (latent JS 0.228, 81st percentile of all pairs; this value survives only in the Phase-IV summary, not in a raw result artifact).

Together: spelled outputs are enough for a weak representational tilt, faster learning and a transient behavioural bias, and not enough for the converged natural line; rare-alias learning is ordinary sample complexity with no symmetry-specific bottleneck. Neither mechanism produces the natural line on its own. Other lexical or learning explanations are not excluded.

APPENDIX F Beyond correspondence

Result 4 could still mean only that both matrices look like the line. This appendix, summarized in §7, asks three narrower questions: whether the corpus conditionals predict behaviour after the explicit geometric and spelling explanations are removed, on held-out rows; whether any such structure depends on the corpus; and whether it appears in a definite order along one pretraining run. The held-out test was pre-registered before any Phase-V comparison was run; its theory baseline was corrected after review (Appendix A, `V4_CORRECTION_REPORT.md`), and the numbers below are the corrected ones. The short answer is that a small and fragile residual survives, and nothing further is established.

F.1 Held-out prediction beyond theory geometry

Objects. $Q^{(m,f)}$: model m's 15×15 behavioural matrix in family f. $C^{(c,F)}$: corpus c's conditional under extraction family F — A: directional 40-word window; B: window with an intervening cue (modulation, chord, key signature, enharmonic, relation); C: high-precision regexes ("from X major to Y major", 317 pairs, too sparse to use); D: document-conditioned, j anywhere later in the document than i. Add-0.5 smoothing, row-normalized; Wikipedia yields 47,247 (A), 10,775 (B), 19,162 (D) ordered pairs from 9,168 key-mentioning documents (the Phase-I helper-word factorization of Result 1 used a wider pass over 14,235 documents containing any of the 42 major- and minor-key spellings; the conditional scan is restricted to the 15 major-key spellings).

Theory features. The corrected v4 baseline distinguishes the true fifths-circle distance d_5 from chromatic cyclic semitone distance. The spelled view also contains absolute and signed open-line displacement, spelling and signature indicators, root/alphabet structure, frequency, and fold-fitted tokenizer irregularity. A merged target is treated as a set of spellings: minimum and maximum line distances and signed interval endpoints replace the former arithmetic centroid, with set-valued orthographic and summed-frequency controls. The rich variant adds prespecified quadratic, interaction and two independent fifth-harmonic terms. Appendix D.6 defines every column.

Tests. (1) Residual correspondence: ridge-regress $\log Q$ and $\log C$ on the corrected features, Spearman of the residuals, joint key-relabeling null ($B = 5{,}000$). (2) Leave-one-source-row-out prediction of Q with theory-only, corpus-only and theory+corpus predictors, scored by within-row Spearman (the pre-registered rank scorer ΔCV), softmax-KL, and within-row R^2; relabeling null ($B = 5{,}000$, $p = (b+1)/(B+1)$) and a document-cluster bootstrap (300 resamples; Appendix D.6). Every continuous predictor, including the corpus column, is standardized from the 14 training source rows inside each fold; the same statistics transform the held-out row, whose response never enters preprocessing. The hold-out is over *source rows*, not templates: four templates are averaged before the primary analysis, and single-template and leave-one-template-out results are reported without selecting a favourable template (Appendix C.7). (3) Enharmonic-twin difference vectors, raw and line-controlled. Two views are retained: the spelled view (15 targets) and a 12-class target-aggregated diagnostic in which scored target probabilities are merged by log-sum-exp and the target features use class sets rather than invented coordinates. This is not the enharmonic-merged behavioural scorer of §2.

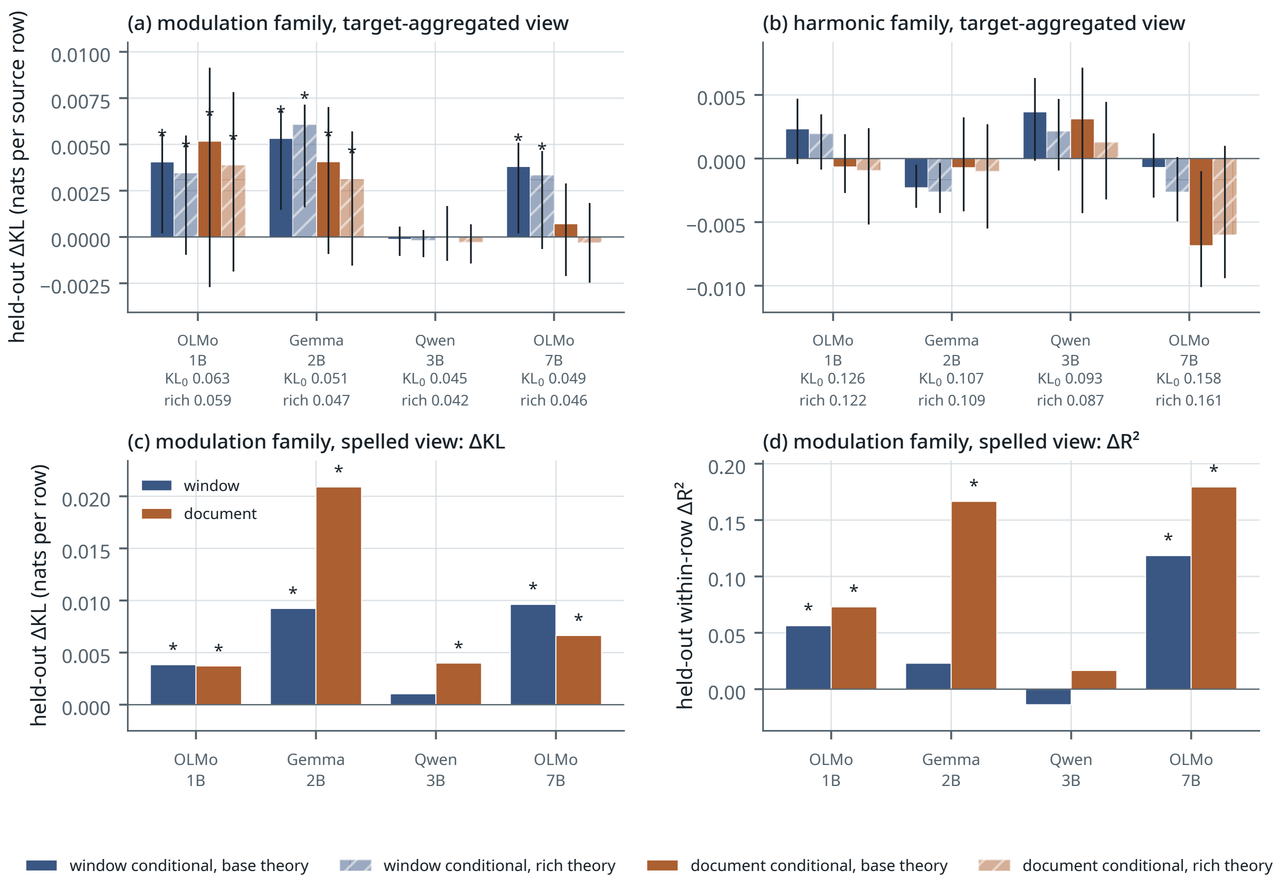

Figure F.1. Held-out gain from the corpus conditionals beyond the theory model, Wikipedia. (a, b) Target-aggregated view: reduction in held-out KL per source row when the corpus row is added to the theory features, for the window (blue) and document (orange) conditionals, with the base (solid) and rich (hatched) theory model; each tick label gives the base and rich theory-only held-out KL, and hatched bars should be read against the rich value; $*$: relabeling $p < .05$; (a) modulation family, (b) harmonic-relation family. (c) Spelled view, modulation family, held-out ΔKL; (d) the same cells scored by within-row ΔR^2 — under the corrected base model the two scorers disagree on significance in one of the eight target-aggregated modulation cells (OLMo-7B × document) and in two of the eight spelled cells. Numbers in Appendix C.

Result 7 — a small and fragile residual

The number. In the target-aggregated modulation view, adding the local-window conditional to the rich theory model reduces held-out KL by +0.0035/+0.0061/−0.0002/+0.0033 nats per row (OLMo-1B/Gemma/Qwen/OLMo-7B; relabeling $p = .034/.0004/.402/.0026$), which is 5.9%/13.0%/−0.4%/7.3% of a theory-only KL of 0.042–0.059. The document conditional gives +0.0039/+0.0031/−0.0003/−0.0003 ($p = .044/.022/.553/.628$). The base theory model gives the same three-versus-one pattern with slightly larger values (window +0.0041/+0.0053/−0.0001/+0.0038, p = .015/.0012/.334/.0030; Table C.2), and a training-row target prior (each target's mean log-probability over the 14 training rows, refitted in every fold) sharpens the relabeling test in the three models that show the effect and leaves Qwen null (rich window +0.0039/+0.0045/−0.0003/+0.0030, p = .0004/.0002/.608/.0010).

Where it comes from. The mean over 15 held-out rows hides that one row does most of the work. C♭ major, a key with 38 mentions in 3.1B words and 48 co-mention events, is the row where the theory model is worst and the corpus helps most: it contributes 66% of OLMo-1B's rich window gain, 44% of Gemma's and 94% of OLMo-7B's. Dropping it leaves +0.0013/+0.0037/−0.0003/+0.0002 (Table C.8). Only Gemma's gain is spread across rows; OLMo-7B's is one row, and Qwen's null flips to +0.0007 if instead C♯ is dropped. The same fragility shows in the estimator: three near-identical corpus predictors give OLMo-1B p = .004, .015 and .107 (Result 4), and the fixed ridge penalty is effectively none (Appendix D.6).

Uncertainty. A document-cluster bootstrap (the 9,168 key-mentioning documents resampled with replacement, count matrices and features rebuilt in each of 300 draws) gives replicate gains shifted toward zero by 22–40%, as expected when a resample with fewer distinct documents weakens a sparse predictor. The percentile intervals therefore exclude zero only for Gemma under the rich model ([+0.0016, +0.0071]; the others are [−0.0009, +0.0054], [−0.0010, +0.0003] and [−0.0006, +0.0046]) and for OLMo-1B, Gemma and OLMo-7B under the base model; the pivotal intervals from the same draws exclude zero for OLMo-1B, Gemma and OLMo-7B under both models (Table C.8). With B = 300 neither rule is precise, and we report both rather than choose. Non-held-out residual correlations follow the same split (window +0.24/+0.28/+0.07/+0.31, p = .019/.0006/.236/.0004; document not significant). The bootstrap was run for the modulation and harmonic-relation families only; in the harmonic family no cell is significant by relabeling, and two cells are significantly *negative* by the base-model percentile interval (Gemma × window −0.0023 [−0.0038, −0.0006], OLMo-7B × document −0.0069 [−0.0100, −0.0011]): adding the corpus row there hurts. The chord family shows nothing either way.

The other view. In the spelled view (15 targets, base model) theory-only KL is already low (0.040–0.072; leave-one-out Spearman 0.88–0.95) because the rows are line-dominated. The gains are +0.0038/+0.0092/+0.0010/+0.0096 for the window conditional (p = .005/.013/.159/.0008) and +0.0037/+0.0209/+0.0040/+0.0066 for the document conditional (p = .006/.0008/.024/.022), with the ΔR^2 scorer significant in five of these eight cells. This view is the stronger one; we lead with the target-aggregated view because it removes target-spelling structure, a choice made after both views had been seen. Note that the aggregated view excludes a source's own pitch class from its targets, so for the six twin sources the enharmonic partner is never scored, and the one relation that motivates the 15-key design is invisible to this test. The complete family, view and extraction grid, null and negative cells included, is in the v4 artifacts and Figure F.1.

Audit note — pre-specification, multiplicity and the two scorers

No single cell of the model × family × extraction grid was pre-specified as primary; the design listed all three context families and competing extractions. All 36 cells of each view remain in the result files. Modulation was emphasized after the v3 grid was seen, and the target-aggregated view was chosen as the lead view after both views were seen; v4 does not promote the document operator because its preview looked favourable. Cell-level relabeling p-values are unadjusted. Counting only the 36 cells of one view and one model, Gemma's rich-window cell survives a Bonferroni factor of 36 ($p = .0004 \times 36 = .014$); counting the 288 Wikipedia held-out tests actually run (36 cells × two views × base/rich × with/without target prior) nothing does (.115). The paper therefore reports the complete grid, the row jackknife and both interval types, and treats replication across models as partial.

The rank scorer ΔCV was pre-registered and the design also named softmax-KL, but only rank was implemented in the first run; KL, and the unregistered within-row R^2 diagnostic, were implemented after the rank result was seen to be near zero. Both remain reported. An earlier hardening pass replaced an under-resolved 300-relabeling estimate b/B with B = 5,000 for Wikipedia, B = 2,000 elsewhere, and $(b+1)/(B+1)$ (Phase-II nulls keep b/B). The v4 correction was more substantive: v3's supposed circle predictor was chromatic cyclic distance, and its target-aggregated line feature was the arithmetic mean of distant enharmonic spellings. V4 separates true fifths-circle and chromatic distances, represents merged line geometry by set extrema and signed endpoints, and refits every scaler inside each source-row fold. These changes reduced the old 14–34% headline to the effects above; the round-5 review then showed that those effects are carried by one row in two models. The KL result should be read as pre-registered in content, post-hoc in timing, corrected in specification, and fragile in composition.

Twin difference vectors ($\log q(\cdot \mid \text{C}\flat) - \log q(\cdot \mid \text{B})$ and so on) correlate strongly with their corpus counterparts for the rarest twin (C♭|B cosine +0.65…+0.92; G♭|F♯ +0.41…+0.86; D♭|C♯ only +0.02…+0.54), but that is the line. After residualizing both on target line position and glyph class, significant alignment survives in 16 of 108 cells (36 spelled-view cells × three twins), scattered across pairs, families and extractions with no consistent cross-model pattern. We do not build on them.

The matched operators as held-out regressors. In the target-aggregated modulation view of this appendix the matched operators no longer separate cleanly. ΔKL in nats per row with the symmetrized conditional is +0.0052, +0.0071, +0.0005 and +0.0069 (p = .0038, .0002, .104, .0002); with the directional conditional +0.0041, +0.0053, −0.0001 and +0.0038 (p = .015, .0012, .334, .0030); with the reverse conditional +0.0019, +0.0065, +0.0004 and +0.0036 (p = .107, .0006, .134, .0044); with the same-count PMI −0.0017, −0.0000, +0.0008 and −0.0022 (p = .964, .267, .049, .999). Three predictors that correlate at $\rho \geq 0.93$ give OLMo-1B one cell that survives a Bonferroni factor of 36, one nominally significant and one null; that spread is the estimator's variance at this effect size, not a ranking of the operators.

F.2 Other corpora

We repeated the corrected base-feature regression on OLMo-Mix-1124 wiki (6,857 key documents, 8,540 window pairs), a 9-shard OLMo-Mix DCLM sample (386 documents, 633 pairs), a post-hoc 54-shard expansion (2,884 documents, 3,843 pairs), and a 2-shard Dolmino DCLM sample (611 documents, 1,320 pairs) (OLMo Team et al., 2025; Li et al., 2024). Dolmino FLAN has only six pairs and is unusable. Each corpus is compared with five independently seeded Wikipedia thinnings matched on pair mass per extraction family. Specificity remains the row-wise difference-in-differences (DiD), (OLMo models − Gemma/Qwen) × (corpus − size-matched Wikipedia), over all nine family×extraction cells and both views. This is exploratory: the corpora are small samples and the 54-shard expansion was added after the first analyses.

Result 8 — neither universality nor a training-data fingerprint

Other corpora do not reproduce the Wikipedia pattern. In the target-aggregated 36-cell grids, 8/0/6/0 cells are significant for OLMo-Mix wiki / 9-shard OLMo-Mix DCLM / Dolmino DCLM / 54-shard OLMo-Mix DCLM (spelled view 7/3/8/2), and every grid has negative cells. The modulation × window row that carries Result 7 is +0.0038/+0.0051/+0.0003/+0.0035 on OLMo-Mix wiki (three significant), about zero on the 9-shard DCLM sample (+0.0000/+0.0001/−0.0007/+0.0015, none significant), Gemma-only on Dolmino (+0.0003/+0.0019/+0.0008/−0.0007) and negative in all four models on the 54-shard expansion (−0.0016/−0.0006/−0.0006/−0.0004). Document rows are no more consistent, and across the five thinning seeds the largest per-cell sample SD is 0.0061 nats (aggregated) and 0.0063 (spelled), sometimes larger than the cell mean. Specificity is directionless: 8 of 72 DiD cells reach unadjusted Wilcoxon $p < .05$, four favouring OLMo and four opposing it (33 positive and 39 negative contrasts; two significant cells of opposite sign in the target-aggregated view, six split three/three in the spelled view; no multiplicity correction is passed). Per-cell values are in Appendix C.3 and Figure C.1. So the corpus conditionals do not generalize the Wikipedia result across corpora, and the specificity test finds no fingerprint. The second statement is weaker than it sounds: Gemma's and Qwen's training data are undisclosed and very likely include Wikipedia and DCLM-like web text, so the OLMo-versus-others contrast could miss a fingerprint that all four models share; the test can reject a large OLMo-specific effect, not the existence of any fingerprint.

The cross-corpus analysis is a failed universality test, not a replacement search. It narrows the positive statement to the corrected Wikipedia cells of Result 7. Even the larger DCLM sample is ≈ 0.1% of DCLM, the paired test has fifteen rows, and corpus-specific frequency features change the theory baseline; those limitations do not convert the null and negative cells into evidence.

F.3 Acquisition along one pretraining run

OLMo-2-0425-1B publishes intermediate checkpoints. We scored eight stage-1 revisions (1B, 21B, 49B, 105B, 294B, 1007B, 1993B, 4001B tokens) and all three stage-2 ingredient endpoints (51B tokens each, same stage-1 start, different seeds) with the behavioural battery, and computed for each the line/circle partials and the target-aggregated residual correspondence with Wikipedia conditionals under the rich theory model (2,000 relabelings, $(b+1)/(B+1)$); Appendix F.2 showed that this residual is not specific to any one corpus, so "residual correspondence" below means correspondence with conditional statistics beyond the theory features.

Result 9 — coarse structure early; residual timing unresolved

The coarse structure appears early. Along stage 1, line | circle rises from −0.24 (1B tokens) to +0.19 (21B), +0.27 (49–294B), +0.34 (1T) and +0.33 (2T, 4T), and reaches +0.42/+0.42/+0.44 at the three stage-2 ingredient endpoints; circle | line is small at most checkpoints apart from a −0.25 excursion at 49B; twin asymmetry falls from 4.3 nats (rare spellings nearly unlearned) to 2.6. The finer residual does not settle. Under the corrected rich baseline the window-residual correlations across stage 1 are +0.01, +0.03, +0.03, +0.05, +0.10, +0.22, +0.10, +0.13, none significant (p = .083 at the 1T peak); the three stage-2 values are +0.21 (p = .023), +0.23 (p = .013) and +0.18 (p = .052); the released model, a different training run and not the trajectory's endpoint (Appendix C.5), gives +0.24 (p = .018); and no document residual is significant at any checkpoint (Table C.5, Figure C.2). Stage 2 raises the line in 3/3 ingredients, but its residual change is not separated from run variation. The earlier reading (first significant at 294B, stable from 4T) is withdrawn.

This is one run and one family with a clean line signal, a residual that peaks at 1T and drops again at 2T for no reason we can name, and no reconstructible data order (none is published for the 1B; only the 32B has official OLMo-core scripts). It is suggestive, not a law.

APPENDIX G Further hidden-state analyses

What remains after controls is small. With the glyph block, commonness, letter identity and alphabet distance partialled out, the best-layer fifths partials are +0.21/+0.29/+0.23/+0.34 (OLMo-1B/Gemma/Qwen/7B). With a max-over-layers correction only the 7B model is significant (p = 0.046 free / 0.018 block-preserving; 1,000 relabelings, $(b+1)/(B+1)$); the other three are at p = 0.09–0.21. The apparent scale curve is three noise-level points and one significant one. Even the 7B signal is line-like (circle | line ≈ 0, line | circle +0.2…+0.3), not an isotropic circle, and it is gone at the final layer. Two further instrument facts come from the 15-key design. First, the *last token* of a multi-token key name is the accidental for two-token keys and the letter otherwise. Probing it produced a line in the key-name geometry of Qwen and 7B (+0.26…+0.37, significant in all six context families under the free null; under the glyph-preserving null Qwen is 0/6 and 7B 3/6) that vanishes under a span-mean probe (line | circle ≈ 0 everywhere). There is no neutral key-token probe when concept names tokenize differently; only the shared prompt-final position is token-uniform. Second, the general aliasing fact: a contiguous block of width $w = 1, \dots, 6$ on the 12-cycle puts 18/37/55/70/80/83% of its energy in the fundamental pair, and root-letter identity by itself puts 34% in the chromatic pair. *Fourier power measures variation with respect to a chosen labelling; it does not identify the feature that caused it.* Karkada et al. (2026)'s Appendix D notes that cyclic and binary attributes can coexist in different spectral sectors; the point here is that a binary attribute can land in the cyclic sector.

At the token-uniform prompt-final position, cross-validated rank regressions show orthography and commonness explaining a cross-validated R^2 of 0.06–0.45 of pair distances per model × family cell (layer means; single layers range from −0.04 to 0.76), with circle or line adding at most 0.03 to any cell's layer mean (single layers exceed 0.04 in 30 of 660 layer-cells, at most +0.10, in OLMo-7B's chord family). OLMo-2-7B has a periodic component at this position in all six context families, relational and non-relational alike (circle | line +0.21…+0.36, layer-corrected free-null p = .004–.03 with 500 relabelings, not corrected across the six families), so it is not a context effect; Gemma has an open one in some families; and cross-fitted "spelling" and "semantic" subspaces overlap heavily (held-out variance shares 0.53–0.76 and 0.46–0.73, summing above one), so separable circle, line and spelling subspaces are not supported. Band-averaged context effects on the periodic component are +0.04…+0.07 in three models but −0.026 in Gemma; only Qwen reaches $p < .05$. Hidden-state geometry is therefore not where this paper's positive claim lives.